

Demand for efficient electric cooking appliances in Ethiopia, Kenya, and Uganda*

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August 31, 2026

Abstract

Urban East African households are widely grid-connected yet cook overwhelmingly with biomass. We elicit willingness to pay (WTP) for an electric pressure cooker, a low-operating-cost electric cooking technology, from 1,306 households in Ethiopia, Kenya, and Uganda using a stated price ladder followed by a binding Becker–DeGroot–Marschak offer. The two elicitation methods bound the subsidy required for 50% adoption at 56–90% of retail price. The bounds are wide because stated WTP overstates revealed demand: 62% of households whose random draw fell at or below their stated maximum refused to buy. Refusal rises steeply with the drawn price conditional on stated WTP, consistent with stated WTP inflated yet ordered by valuation. Refusals track the purchase price rather than the appliance’s 2–8-month payback, and retail prices equal 47–67% of monthly expenditure, consistent with liquidity constraints. Randomized operating-cost information does not raise WTP (−0.74 percentage points, SE 1.34).

*Corresponding author: Erin Litzow (erin.litzow@utdallas.edu). The authors declare they have no conflicts of interest. We gratefully acknowledge financial support from Environment for Development (EfD). We thank attendees at the 2023 EfD Annual Meeting, attendees of the 2025 AERE annual meeting, attendees of the 2025 SWEEEP, and seminar participants at Duke University and the the University of British Columbia for comments and feedback. All errors are our own.

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1 Introduction

Clean-cooking adoption in Sub-Saharan Africa lags far behind electricity access. National electricity access in 2023 reached 76.2% in Kenya, 55.4% in Ethiopia, and 51.5% in Uganda, but clean-cooking access in the same year was 31.6%, 7.1%, and 1.8% (IEA, IRENA, UNSD, World Bank, & WHO, 2025). Cooking with biomass, the dominant alternative, is a leading source of household air pollution (Pattanayak, Pakhtigian, & Litzow, 2018; Jeuland et al., 2021; Jeuland, Pattanayak, & Bluffstone, 2015) and consumes hours of daily labor that fall disproportionately on women and children (Krishnapriya, Chandrasekaran, Jeuland, & Pattanayak, 2021). Electrifying cooking is technically attractive in our three settings because grid generation is 85–97% renewable (International Energy Agency, 2025): a household switching from biomass displaces solid fuel with predominantly low-carbon power. The electric pressure cooker (EPC) is the cleanest, fastest, and lowest-operating-cost electric cooking option for typical urban dishes. Per-meal energy use is roughly half that of an open coil, and per-meal operating cost is below charcoal and LPG at published residential tariffs (Lakew Tesema, Tesfaye, Paulos, Tesfamichael, & Mulugetta, 2020; Leary et al., 2019; Naluwagga et al., 2022). Yet EPC ownership in all three countries remains negligible. Even in Kenya, the region’s most developed eCooking market, under 3% of the population owns any electric cooking appliance, fewer than 1% cook mainly with electricity, and cumulative EPC sales as of 2021 were measured in thousands (Leary et al., 2022; Bolo et al., 2023). In Uganda, 0.8% of households cook primarily with clean fuels (Energy Sector Management Assistance Program (ESMAP), 2025). Electric cooking is widespread only in urban Ethiopia, where 15.3% of households use a clean-fuel stove, mostly conventional electric stoves rather than energy-efficient appliances (Energy Sector Management Assistance Program (ESMAP), 2018). We ask three questions about urban household demand for electric cooking. First, at what subsidy level do urban grid-connected households in Ethiopia, Kenya, and Uganda buy an EPC? Second, how far does stated willingness to pay overstate demand revealed by real-stakes purchase decisions, and can that gap be diagnosed and bounded? Third,

what explains low revealed demand for a technology whose fuel savings repay its purchase price within months? A randomized cost-information module provides an embedded test of one candidate explanation. Regulators, utility staff, and clean-cooking stakeholders repeatedly report that perceived operating cost is the binding consumer barrier (Leary & Onjala, 2022; Kersey et al., 2024). Consistent with this, 47.3% of our respondents identify electricity as the costliest fuel for cooking a staple dish before treatment, plausibly accurate for resistive coil stoves but not for the EPC.

We survey 1,306 urban grid-connected households (501 in Addis Ababa, 537 in Nairobi and Kiambu County, and 268 in Kampala), fielded September through December 2024. We elicit willingness to pay (WTP) for an EPC using a Becker–DeGroot–Marschak (BDM) mechanism (Becker, DeGroot, & Marschak, 1964; Berry, Fischer, & Guiteras, 2020). The respondent reports the highest price tier she would accept on a country-specific 10–11 point ladder running from 0 to 100% of the local market price, then physically draws a binding offer from an opaque bag and purchases the EPC if and only if the drawn price falls at or below her stated maximum. This within-subject design yields both a full stated demand ladder and one incentive-compatible purchase decision for every respondent. Earlier in the survey, each respondent completes a parallel stated price ladder for a low-value consumer good (a jar of jelly), which we use as a placebo measure of generic ladder response style. We randomly assigned half of respondents, at the household level, to receive a cost-information module immediately before the BDM. The module presented per-meal operating-cost comparisons across the EPC and the locally relevant alternatives, drawn from the MECS Kitchen Lab and country eCookbook studies (Lakew Tesema et al., 2020; Leary et al., 2019; Naluwagga et al., 2022). We cluster standard errors at the sampling-cluster level (63 clusters) and report wild-cluster bootstrap inference (Cameron, Gelbach, & Miller, 2008; MacKinnon & Webb, 2018; Roodman, Nielsen, MacKinnon, & Webb, 2019).

Mean WTP is far below market price, and reaching adoption targets requires deep subsidies. Pooled mean stated (ladder) WTP is 33% of market price. Country means are 17% in

Ethiopia, 43% in Kenya, and 40% in Uganda, or \$14, \$36, and \$29 at the market reference prices used in the elicitation. The revealed-preference demand curve, traced from binding BDM purchase decisions, is steeper still. Moving half of urban households across the purchase threshold requires subsidies of 90.5% in Ethiopia, 81.7% in Kenya, and 89.0% in Uganda. These are intention-to-treat figures. Households never offered a draw because they stated zero WTP, 114 in Ethiopia, count as non-buyers at every positive price. The stated ladder implies 81.5%, 56.2%, and 60.5%, understating the required transfer by 9 to 29 percentage points. We argue that the two elicitations bound the demand curve under transparent assumptions. We construct interior estimates by reweighting stated responses by each household’s predicted overstatement probability, or by capping stated WTP at refused draw prices. Both approaches land between the bounds. Either way, these magnitudes exceed any subsidy currently committed in the three settings ([Kenya Power and Lighting Company, 2024](#); [Stockholm Environment Institute, Berkeley Air Monitoring Group, Clean Cooking Alliance, & CIRCODU, 2025](#); [World Bank, 2023](#)). Ethiopian retail prices of \$130–192 are two to four times the Kenyan (\$35–53) and Ugandan (\$62–94) levels, which pushes adoption further out of reach there despite similar dollar WTP in Kenya and Uganda.

The gap between elicitations is structured, not noise. Among the 340 households whose randomly drawn price fell at or below their stated maximum, 62.1% refused the binding offer. Refusal rises from 3.9% at near-free draws to 100% at the top tier and, holding stated WTP fixed, still rises 1.25 percentage points per point of drawn price (wild-cluster bootstrap $p < 0.001$), while a higher stated maximum predicts compliance at any given price. Strategic overstatement aimed at preserving a cheap option predicts bunching at the top ladder tier, which only 0.4–7.8% of respondents choose. We interpret the pattern as stated WTP that is inflated in level yet positively correlated with underlying valuation, though we cannot rule out subtler strategic responses.

Overstatement behaves like a general survey response style rather than an EPC-specific belief: high stated shares on the placebo jelly ladder and fewer years of schooling predict refusal of

the EPC draw, while enumerator-rated BDM comprehension does not. Overall predictability is nonetheless weak (pseudo- R^2 of 0.08), so household-level correction of stated WTP is infeasible in these data, which is why we report bounds rather than a single corrected demand curve.

Low revealed demand is not explained by poor economics. At published residential tariffs, median fuel savings repay the drawn EPC price in 2.1–4.8 months and the full market price in 4.7–7.8 months, equivalent to breaking even after 253–495 meals against roughly 3,650 meals over the appliance’s life. Refusal tracks the sticker price, not the return. Conditional on the drawn price, predicted payback does not predict refusal ($p = 0.88$). Two facts point instead to liquidity constraints and non-salient savings. The market price equals 47–67% of median monthly household expenditure across the three countries, and roughly 70% of respondents expect zero fuel-cost savings from the EPC. Consistent with the salience margin being hard to move, the randomized cost-information module shifts stated WTP by -0.74 percentage points (standard error 1.34, Table 3). The 95% confidence interval rules out positive effects larger than $+1.9$ percentage points. These results align with evidence that credit, rather than information, binds adoption of high-return energy durables (Berkouwer & Dean, 2022; Filippini, Kumar, & Srinivasan, 2026; Mahadevan, Martinez, McCord, Meeks, & Pradhananga, 2025).

This paper makes four contributions. First, we provide, to our knowledge, the first multi-country revealed-preference estimates of EPC demand in Sub-Saharan Africa. These estimates speak to a central puzzle in environmental economics. Willingness to pay for environmental goods with high private returns is low in low-income settings (Greenstone & Jack, 2015). The EPC is a stark instance. Per-meal operating costs are below charcoal and LPG, efficiency gains are engineering-verified, and adoption is near zero. The closest priors elicit WTP for related clean-energy or clean-water durables in single-country LMIC settings (Berkouwer & Dean, 2022; Berry et al., 2020; Mahadevan et al., 2025; Pattanayak et al., 2019), and existing EPC-specific evidence is stated-preference or drawn from small post-pilot exercises.¹

¹Bawakyillenuo, Agbelie, Crentsil, and Danquah (2025) survey 1,203 urban and peri-urban Accra households after video-documentary cooking-cost demonstrations and report stated WTP below EPC market price. MECS-program pilots in Tanzania and Rwanda run end-of-pilot WTP exercises with 51–100 mini-grid households (Kweka et al., 2021; Shuma et al., 2022; Bisaga & Menyeh, 2022). None of these elicits

Second, we develop a diagnose-and-bound approach to hypothetical bias in stated demand data. The within-subject design pairs a full stated ladder with one binding draw for every respondent, which identifies the refusal-price gradient that separates flat hypothetical bias from inflation correlated with true valuation. The gradient brings to field data the stated-preference literature’s core insight. Survey responses must be interpreted through the incentive structure respondents perceive (Carson, Groves, & List, 2014; Vossler & Zawojksa, 2020). The placebo ladder for an unrelated low-value good adds a portable diagnostic for ladder response style at a cost of minutes of survey time. A callback window for delayed payment makes a same-day liquidity shock an unlikely explanation for refusals, and restricting to high-comprehension respondents, the consistency correction recommended by Berry et al. (2020), leaves both demand curves essentially unchanged. Field-BDM studies of smaller goods report smaller ladder-binding gaps (Berry et al., 2020; Berkouwer & Dean, 2022). Our gap of 9–29 percentage points in subsidy terms shows that stated ladders for lumpy durables can materially misstate the transfer needed to reach an adoption target. However, bounds are the honest product of this design. The overstatement we detect is only weakly predictable at the household level, so corrections that classify and remove overstaters do not travel. This caution is consistent with experimental evidence that ex post adjustments to stated choices, such as certainty recoding, distort demand estimates when decisions are consequential (Chopra & Vossler, 2026).

Third, we extend the clean-cooking-adoption literature across three countries at distinct stages of fuel transition. Monthly expenditure and prior EPC familiarity predict WTP within country, baseline cooking fuel and grid access also predict WTP, and outage exposure tracks WTP across countries but not within (Lewis & Pattanayak, 2012; Jeuland et al., 2021; Pattanayak et al., 2018). Kenya, mid-transition with 41.6% of urban households on LPG or electricity as the primary cooking fuel (Energy Sector Management Assistance Program (ESMAP), 2019), shows higher WTP than Uganda, where 98% of urban households remain at MTF cooking Tier 1 or below (Energy Sector Management Assistance Program (ESMAP), 2025), and Ethiopia,

incentive-compatible WTP, and none spans multiple countries. The nearest experimental evidence is an RCT randomizing free EPC offers in Goma, DRC (Desbureaux et al., 2025; Cikesa, Collart, Desbureaux, Stoop, & Verpoorten, 2026), which studies usage and welfare conditional on receipt rather than demand.

where households face 19.78 outage hours per week versus 5.56 in Kenya and 5.87 in Uganda. The level and slope of these correlates shift across transition stages, consistent with [Guta et al. \(2022\)](#), [Gill-Wiehl, Ray, Katikiro, Kammen, and Hubbard \(2025\)](#), and [Pattanayak et al. \(2019\)](#). The null effect of our per-meal cost-information treatment tests the information explanation for the low-WTP puzzle directly ([Greenstone & Jack, 2015](#)). Correcting a misperception held by 47.3% of respondents moves WTP by at most +1.9 percentage points. Alongside parallel nulls in [Filippini et al. \(2026\)](#), [Berkouwer and Dean \(2022\)](#), and [Mahadevan et al. \(2025\)](#), this yields a design lesson. Short-horizon operating-cost framings do not move clean-energy-durable demand, even though persuasive appeals raise take-up of low-cost efficient lightbulbs by 6 percentage points in Brazilian favelas ([Toledo, 2016](#)).

Fourth, our estimates speak directly to current clean-cooking policy. Kenya, Uganda, and Ethiopia have all announced electric-cooking transition targets ([Kenya Power and Lighting Company, 2024](#); [Stockholm Environment Institute et al., 2025](#); [World Bank, 2023](#)), but none subsidizes EPC purchase at scale. Our bounds identify the order of magnitude required, and a program that budgets from stated-preference data alone would undershoot the required transfer by up to 29 percentage points, so the bounds themselves are the policy-relevant object. The mechanism evidence points the same way. Payback occurs within months, refusal tracks the purchase price rather than the return, and prices sit near half of monthly expenditure. Together these suggest that financing instruments in the spirit of [Berkouwer and Dean \(2022\)](#) are the natural next lever to test, rather than further information. Our design cannot resolve the credit question, but the size of the subsidy gap makes it a time-sensitive one.

2 Context

2.1 Energy access

Electric cooking requires a connection to high-powered, usually grid, electricity.² National electricity access rates in 2023 were 76.2% in Kenya, 55.4% in Ethiopia, and 51.5% in Uganda (IEA et al., 2025). Urban rates were uniformly higher, at 96.0% in Kenya, 94.7% in Ethiopia, and 76.4% in Uganda.³ However, these headline figures apply the Multi-Tier Framework (MTF) Tier 1 threshold, which corresponds only to enough power for lighting and phone charging (12–200 Wh/day). EPC operation requires roughly Tier 4 service, meaning a 1 kW peak draw for 30–60 minutes with at least eight hours of daily availability spanning the evening cooking window.

Tier-resolved estimates from the World Bank’s Energy Sector Management Assistance Program (ESMAP) MTF Diagnostic Surveys show that the urban populations we sample sit well above the Tier 1 floor. In Addis Ababa, 91.1% of households are at Tiers 3 to 5 (Energy Sector Management Assistance Program (ESMAP), 2018), 37.9% of grid-connected Kenyan households are at Tier 3 or higher (Energy Sector Management Assistance Program (ESMAP), 2019), and most grid-connected urban Ugandan households fall at Tier 4 (Energy Sector Management Assistance Program (ESMAP), 2025). Reliability, rather than the connection itself, is the binding operating margin. More than half of grid-connected Kenyan households experience at least three outages per week (Energy Sector Management Assistance Program (ESMAP), 2019), 57.6% of electrified Ethiopian households face 4–14 disruptions per week (Energy Sector Management Assistance Program (ESMAP), 2018), and Ugandan grid-connected households consume on average only 42 kWh per month, well below the level implied by daily electric cooking (Energy Sector Management Assistance Program (ESMAP), 2025).

²Off-grid solar PV with battery storage can in principle power an EPC, but a configuration sufficient for a full cooking cycle, roughly 1 kW peak PV with 2–3 kWh of battery storage, is far larger and more expensive than the solar home systems commonly deployed in our study countries. Solar thermal cookers exist but cannot replicate the speed or pressure of an EPC.

³Country-by-urban figures drawn from the underlying SDG7 tracking database at trackingsdg7.esmap.org/downloads, which supports the country totals reported in the printed report.

Clean-cooking access lags far behind electricity access, even where the grid is dense. National rates of access to clean cooking are 31.6% in Kenya, 7.1% in Ethiopia, and 1.8% in Uganda (IEA et al., 2025). Urban cooking-fuel mixes differ sharply across the three countries, and place each country at a distinct stage of fuel transition.

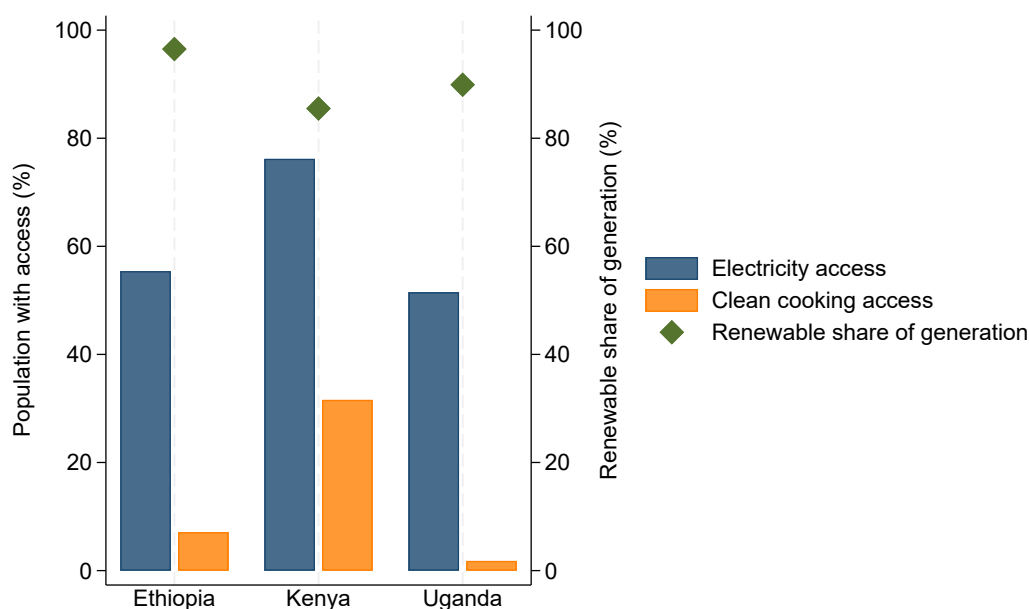
In urban Kenya, 41.6% of households use LPG or electric stoves as their primary cooking fuel (Energy Sector Management Assistance Program (ESMAP), 2019), placing Kenya in the middle of a fuel transition already underway. In urban Uganda, by contrast, 62% of households cook with charcoal and 32% with wood, and 98% of urban households remain at MTF cooking Tier 1 or below (Energy Sector Management Assistance Program (ESMAP), 2025). In Ethiopia, only 15.3% of urban households nationally use an electric stove as their primary cooking technology (Energy Sector Management Assistance Program (ESMAP), 2018), although our Addis Ababa sample falls within an EEU service area where the underlying electricity tier supports electric cooking. This pattern of high urban electricity access alongside low clean-cooking access, with country-specific transition stages, motivates our urban sampling frame and our cross-country comparison.

2.2 Electricity sector

Renewables supply 85–97% of generation across the three countries. Ethiopia’s grid is 96.5% renewable, almost entirely hydropower. Uganda’s is 89.9% hydropower. Kenya’s is 85.5% renewable, split across geothermal (43.2%), hydropower (28.3%), and wind (14.0%) (International Energy Agency, 2025). A household that switches from charcoal or fuelwood to an EPC therefore displaces solid biomass with predominantly low-carbon electricity rather than substituting one carbon-emitting source for another. However, marginal generation at peak demand can rely on thermal backup, so the carbon intensity of EPC use at peak hours can exceed the average implied by the renewable shares reported here.

Domestic electricity is priced under an increasing block tariff (IBT) in all three countries. For an urban household consuming 50–150 kWh per month, the range that brackets EPC

Figure 1: Electricity access far exceeds clean-cooking access in Ethiopia, Kenya, and Uganda, and grid generation is predominantly renewable



Notes: Left axis (bars): national electricity-access and clean-cooking-access rates for 2023 from [IEA et al. \(2025\)](#). Electricity access follows the Multi-Tier Framework Tier 1 (IEA “basic bundle”) threshold for household electricity service. Clean-cooking access is the share of the population relying primarily on clean fuels and technologies (electricity, LPG, natural gas, biogas, solar, alcohol) as defined by WHO (2014) normative technical recommendations. Right axis (diamond markers): renewable share of electricity generation from [International Energy Agency \(2025\)](#). Reference years and composition: Ethiopia 96.5% (hydro), 2023; Kenya 85.5% (geothermal 43.2%, hydro 28.3%, wind 14.0%), 2024; Uganda 89.9% (hydro), 2023.

use, the modal block tariff is 16.54 KES/kWh in Kenya, 756.2 UGX/kWh in Uganda, and 0.7670 ETB/kWh in Ethiopia for postpaid residential customers.⁴ Converted to USD, these correspond to approximately \$0.13/kWh in Kenya (rising to \$0.19/kWh all-in once fuel-cost adjustments, levies, and VAT are applied), \$0.20/kWh in Uganda, and \$0.006/kWh in Ethiopia.⁵ Uganda’s schedule is distinctive in one respect. Above 80 kWh per month, the regulator applies a reduced “cooking tariff” of 412.0 UGX/kWh (\$0.11/kWh) up to 150 kWh, designed to lower the marginal cost of electric cooking. The Ethiopian residential tariff is heavily subsidized below 200 kWh per month and is more than an order of magnitude lower than Kenyan or Ugandan rates.

Metering composition differs across our samples and matters for Ethiopia’s tariff schedule but not for the other two. In our data, 94% of Ugandan households and 70% of Kenyan households use prepaid meters, topping up a meter in advance rather than receiving a monthly bill, while 99% of the Ethiopian sample is on postpaid billing.⁶

Kenya and Uganda are actively pursuing e-cooking adoption via national policy. Kenya Power’s Pika na Power initiative targets 500,000 households for transition to electric cooking (Kenya Power and Lighting Company, 2024), and Uganda’s draft National Integrated Clean Cooking Strategy (September 2025) targets 18% of households cooking primarily with electricity by 2040, supported by the reduced cooking tariff and World Bank–financed cookstove lending under the Electricity Access Scale-up Project (Stockholm Environment Institute et al., 2025; World Bank, 2021; Uganda Energy Credit Capitalisation Company, 2024; MECS Programme, 2024). Ethiopia’s updated NDC calls for a shift from unsustainable biomass demand to electric stoves, but the target is qualitative, and the World Bank–supported

⁴Sources: Energy and Petroleum Regulatory Authority (EPRA) retail tariff schedule, Kenya, <https://www.epra.go.ke/>; Electricity Regulatory Authority (ERA) approved tariffs Q1 2026, Uganda, <https://www.era.go.ug/>; Ethiopian Electric Utility postpaid residential tariff, <https://www.eeu.gov.et/contents/postpaid>.

⁵USD figures use late-2024 average exchange rates of approximately 129 KES/USD, 3,700 UGX/USD, and 127 ETB/USD, corresponding to the period of our data collection. The Ethiopian birr was floated in July 2024 and the ETB/USD rate roughly doubled in the months immediately preceding our Ethiopia survey wave. The USD-equivalent tariff reported here therefore already reflects the post-devaluation rate.

⁶The EEU prepaid schedule is uniformly lower across consumption blocks. Ugandan and Kenyan retail tariffs do not differ by meter type.

Ethiopia Electrification Program expands grid access without directly targeting clean cooking (IRENA, 2023; World Bank, 2023). None of these frameworks currently subsidizes EPC purchase at scale. Our Section 5 estimates identify the subsidy level required to close the gap between observed willingness to pay and retail prices.

2.3 The electric pressure cooker (EPC)

The electric pressure cooker (EPC) is an insulated, thermostatically controlled, pressurized electric cooking appliance. A standard 6-liter household EPC draws roughly 1 kW at peak and consumes 0.3–0.7 kWh per meal, with cooking times of 20–40 minutes (Lakew Tesema et al., 2020; Leary et al., 2019; Naluwagga et al., 2022). The insulated pressurized chamber retains heat and shortens cooking time, so per-meal energy use is roughly half that of an open-coil hotplate cooking equivalent dishes (Lakew Tesema et al., 2020; Leary et al., 2019; Naluwagga et al., 2022). At published residential tariffs in Ethiopia, Kenya, and Uganda, per-meal EPC operating cost is comparable to LPG and below per-meal charcoal cost for typical urban dishes (Lakew Tesema et al., 2020; Leary et al., 2019; Naluwagga et al., 2022). This cost gap is the central object of the information treatment described in Section 4.4. Appendix Figure D1 shows representative models sold in the three countries.

We characterize EPC supply through semi-structured interviews with a small non-random sample of retailers and importers in each capital.⁷ Retail prices for 5–7 L units differ by roughly a factor of four across countries, at \$35–53 in Kenya, \$62–94 in Uganda, and \$130–192 in Ethiopia. The gap is consistent with domestic assembly in Kenya versus import dependence plus post-devaluation pass-through in Ethiopia. Consumer financing is rare (only two interviewed Ugandan firms offer installment plans), and sellers in all three countries identify low household demand as the binding constraint on sales, attributing it to high appliance cost, customer concerns about electricity bills, and limited consumer awareness rather than to supply-side bottlenecks in stock, imports, or service capacity.

⁷Authors’ semi-structured interviews with EPC retailers and importers in Addis Ababa, Nairobi, and Kampala, 2024.

3 Sampling and Data

3.1 Sampling

Our analysis sample is 1,306 urban households across the three countries (501 in Ethiopia, 537 in Kenya, and 268 in Uganda). Sampling in each country followed a stratified multi-stage design that produced clustered samples of urban, grid-connected households. Refusals were rare in all three countries. Ethiopia recorded a single refusal, and in Kenya and Uganda field protocols replaced the few refusing or ineligible households with the next eligible household.⁸

In Ethiopia, we drew the sample from the Ethiopian Electric Utility (EEU) customer list in Addis Ababa, using the four EEU service districts (North, South, East, West) as strata. Within each district we selected two service centers, two customer books per center, and 31–32 households per book by systematic random sampling, for a target of 500.⁹

In Kenya, we purposively selected seven peri-urban sub-counties with high biomass-cooking shares in Kiambu County (Kikuyu, Kiambaa, Ruiru, Kabete, and Kiambu township) and Nairobi County (Dagoreti North and Dagoreti South). Within each sub-county we randomly drew three wards, then sampled households along a four-household skip interval starting from a randomly chosen first household. Eligibility required a working grid connection and no existing EPC. We replaced ineligible households with the next household in the walking direction. The field team implemented the no-EPC screener strictly, so the Kenya sample excludes existing EPC adopters by construction, while the Ethiopia and Uganda samples do not. This mechanically truncates the top of the Kenyan WTP distribution and should be kept in mind when comparing demand across countries. However, the

⁸Field teams did not closely track refusals and replacements. In Uganda, 268 of the 288 sampled households completed the main survey. Households that refused or could not be reached were replaced where possible.

⁹EEU customer books are meter-reading routes defined by the utility’s infrastructure network, not by government administrative boundaries. One book can span multiple administrative-level-3 districts, and one district can contain several books. Book identifiers were not retained in the survey data, so our Ethiopia cluster identifier is the administrative-level-3 district (13 clusters), which does not map one-to-one to the 16 sampled books.

screened sample is arguably the policy-relevant population for programs such as Pika na Power that target households not yet cooking with electricity.

In Uganda, we sampled from two divisions of Kampala (Kawempe and Rubaga), randomly drawing 12 parishes per division and two villages per parish. Within each village we listed 12 households at pre-specified positions along secondary roads on both sides of the village’s primary road, replacing households without a grid connection with the immediately adjacent household (574 households listed in total). We then randomly selected six listed households per village for interview, for a target sample of 288, of which 268 interviews were completed. Uganda’s sample is roughly half the size of Ethiopia’s and Kenya’s because budget constraints capped the number of households we could enroll there.

3.2 Survey instrument

The questionnaire is a structured, enumerator-administered household interview. [Appendix A](#) describes the BDM script and elicitation protocol in more detail.

The instrument opens with consent, a screener (grid connection in all three countries, plus no existing EPC in Kenya), and demographics for the respondent, household head, and primary cook, together with a household roster by sex and four age bins. We then collect a cooking-energy module covering, for each fuel and stove the household uses, source, price, frequency of use, and associated time and labor, alongside perceived relative fuel costs for a representative dish and the household’s grid source, outage frequency, and outage duration.

Next come the information treatment and the BDM elicitations. We first administer an incentivized practice BDM for a low-stakes, locally familiar consumer item. The item is petroleum jelly in Uganda, a bar of soap in Kenya, and a 1-liter bottle of liquid soap in Ethiopia. We refer to this practice elicitation as the jelly BDM throughout the paper, following the East African usage for the household cosmetic. Treated respondents then receive the information module ([§4.4](#)) before the focal EPC BDM ([§4.1](#)). Control respondents proceed directly to the EPC BDM. After the EPC BDM we record whether the

respondent wished to revise their bid, any objection they raised, and an enumerator-rated comprehension score that we use in robustness checks.

The survey closes with mediator and ability-to-pay measures. We record women’s freedom of movement, group membership, and bank-account ownership. We then collect last-month expenditure across nine categories, a 25-item asset inventory (including a non-electric pressure cooker), recent shocks, and food security. We close by eliciting respondent risk and time preferences through hypothetical scenarios. The enumerator then briefly rates respondent comprehension and the household’s relative economic status.

We authored the instrument in English and administered it orally in local languages. Bilingual enumerators rendered the questions into Amharic, Swahili, or Luganda in real time, and each team standardized delivery through intensive training and mock interviews in the survey languages. Appendix A (§A.8) documents the procedure by country.

3.3 Data collection

We piloted the instrument in each of the three study sites in July 2024, finalized the questionnaire and BDM script in August 2024, and trained enumerators in the weeks immediately preceding each country’s fieldwork. We staggered fielding across countries between September and December 2024. Uganda fielded in September 2024, Kenya in October–November 2024, and Ethiopia in November–December 2024. Country teams comprised 10 enumerators in Ethiopia, 10 in Kenya, and 12 in Uganda, each supervised by an in-country field manager.

Enumerators administered the survey face-to-face on tablets using KoboToolbox, in Amharic in Ethiopia, in Swahili or English in Kenya, and in Uganda mainly in Luganda, with a small number of interviews in English.

We monitored fieldwork through three procedures. First, field-management teams ran high-frequency checks on incoming submissions to flag enumerator-level outliers, skip-pattern violations, and implausible values, and resolved flags with the enumerator before the next

day of work. Second, supervisors and enumerators back-checked completed interviews by re-contacting respondents to verify key questions. Ethiopia back-checked all households that received an EPC plus a random subset of other households. Third, enumerators captured GPS coordinates at each interview location to support spot-checks and to confirm that sampled households were enumerated. Country teams layered additional oversight on these procedures. In Ethiopia, researchers made random unannounced field visits and gave enumerators real-time feedback by phone. In Uganda, enumerators worked in pairs under a team leader who reviewed each day’s interviews.

3.4 Sample characteristics

Table 1 reports respondent and household characteristics by country. Respondents are mostly women (87% pooled), the primary cook is almost always a woman (96%), and 90% of respondents are the household head or spouse, a senior household decision-maker in nearly all cases. Female-headed households make up 37% of the pooled sample. We surveyed during weekday daytime hours, which raised the female share of respondents because women were more likely to be at home.

Awareness of electric pressure cookers is low and varies sharply by country. Roughly one in three respondents (34% pooled) reports familiarity with EPCs, from 19% in Ethiopia to 52% in Kenya. We measured familiarity at the start of the survey, before any EPC-specific module, to avoid priming respondents.

Respondents are predominantly impatient and variance-averse. They chose immediate over deferred payment in 4.23 of six time-preference scenarios on average, and the variable income stream over the stable one in 1.63 of six variance-tolerance scenarios. However, the two counts are essentially uncorrelated in the pooled sample ($\rho = -0.02$, $p = 0.49$), so impatience and variance tolerance identify distinct subpopulations rather than a single low-discount, low-variance type. We test below whether either measure predicts WTP within country.

Table 1: Respondent and household characteristics, by country

	Ethiopia	Kenya	Uganda	All
Respondent female	0.83 (0.37)	0.92 (0.27)	0.86 (0.35)	0.87 (0.33)
Respondent age	45.93 (15.40)	42.98 (13.96)	43.29 (15.74)	44.18 (14.95)
Respondent is HH head or spouse	0.84 (0.37)	0.95 (0.21)	0.90 (0.30)	0.90 (0.30)
Impatience score (# today, 0–6)	3.88 (2.21)	4.44 (2.16)	4.46 (1.94)	4.23 (2.15)
Variance tolerance (# variable, 0–6)	1.25 (1.46)	1.72 (1.96)	2.15 (1.96)	1.63 (1.82)
HH head female	0.39 (0.49)	0.32 (0.47)	0.44 (0.50)	0.37 (0.48)
Household size	5.14 (2.14)	4.26 (1.70)	5.16 (2.71)	4.78 (2.15)
Primary cook female	0.99 (0.12)	0.96 (0.19)	0.93 (0.26)	0.96 (0.19)
Primary cook age	39.83 (15.00)	41.72 (13.65)	38.05 (14.59)	40.24 (14.43)
Primary cook is spouse of HH head	0.40 (0.49)	0.63 (0.48)	0.41 (0.49)	0.49 (0.50)
Monthly expenditure (USD)	159.15 (141.19)	250.71 (494.03)	237.18 (510.16)	212.81 (403.67)
Familiar with EPC	0.19 (0.39)	0.52 (0.50)	0.24 (0.43)	0.34 (0.47)
Cooks with charcoal	0.94 (0.24)	0.62 (0.49)	0.94 (0.23)	0.81 (0.39)
Cooks with LPG	0.11 (0.31)	0.93 (0.25)	0.19 (0.40)	0.46 (0.50)
Cooks with electricity	0.95 (0.21)	0.04 (0.19)	0.31 (0.46)	0.44 (0.50)
Cooks with firewood	0.22 (0.42)	0.26 (0.44)	0.30 (0.46)	0.25 (0.44)
Grid is main elec. source	1.00 (0.00)	0.90 (0.30)	0.91 (0.29)	0.94 (0.23)
Typical outage hrs/week	19.78 (17.15)	5.56 (8.70)	5.87 (8.40)	11.06 (14.32)
Observations	501	537	268	1,306

Notes: Cell entries report variable means by country, with standard deviations reported in parentheses. The bottom row reports the number of households in each column. Sample is the analysis sample across Ethiopia, Kenya, and Uganda. Uganda’s smaller sample reflects budgetary restrictions that capped enrollment relative to Ethiopia and Kenya. Monthly expenditure is converted to USD at survey-period exchange rates. “Cooks with [fuel]” indicators equal one if the household uses the fuel for any cooking; the four fuel indicators are not mutually exclusive, so their column sums can exceed one (stove stacking). Typical outage hours per week is the self-reported total duration of all outages in a typical week, winsorized at country-specific 95th percentiles (caps of 63 hours/week in Ethiopia, 30 hours/week in Kenya and Uganda) to limit influence from implausible right-tail responses; households reporting zero outages were not asked the duration question and are coded as zero hours.

Cooking-fuel mixes place each country at a distinct stage of fuel transition. In the pooled sample, 81% of households cook with charcoal for any meal, 46% with LPG, 44% with electricity, and 25% with firewood. The indicators are not mutually exclusive, so a single household can be counted in multiple columns (stove stacking is common). The primary stove, the single fuel each household names as its main cooking technology, divides the countries starkly. Ethiopia is overwhelmingly main-fuel electric (76%, typically an electric mitad for injera), Kenya is overwhelmingly main-fuel LPG (81%), and Uganda is overwhelmingly main-fuel charcoal (77%). Kenya has transitioned to a clean recurrent-cost fuel, Ethiopia uses its near-universal grid for primary cooking but layers charcoal on top, and Uganda remains anchored to solid-biomass primary cooking despite high grid access.

Grid connectivity is near-universal. In the pooled sample, 94% of households report the main national grid as their primary electricity source, and Ethiopia’s sample is uniformly main-grid connected. EPCs are therefore technically operable for nearly all sampled households, and the cross-country differences in WTP we report below are not driven by differential access to electricity. Outage exposure is high but uneven. Ethiopian households report 19.8 hours of weekly outage time on average, compared with 5.6 hours in Kenya and 5.9 hours in Uganda (self-reported hours in a typical week, winsorized at country-specific 95th percentiles; see table notes), consistent with ESMAP MTF diagnostics ([Energy Sector Management Assistance Program \(ESMAP\)](#), 2018, 2019). Outage exposure measures the reliability margin that determines whether the grid can support a full evening cooking cycle on an EPC, and we test below whether reliability predicts WTP.

Three nested samples organize the analysis. The full sample of 1,306 households answers the stated ladder. The per-protocol BDM sample completes a binding draw, with 387 households in Ethiopia (excluding the 114 gated ladder-zero respondents), 536 in Kenya, and 264 in Uganda. The at-risk sample comprises the 340 households whose draw fell at or below their stated maximum, so that the commitment rule required purchase (75 in Ethiopia, 159 in Kenya, and 106 in Uganda). Within the at-risk sample, 211 households (62.1%) refused the binding

offer (33 of 75 in Ethiopia, 115 of 159 in Kenya, and 63 of 106 in Uganda). These refusals are the raw material for the diagnostics of §4.2.

4 Empirical Strategy

4.1 Eliciting willingness to pay

We elicit each respondent’s willingness to pay (WTP) for an electric pressure cooker using a Becker–DeGroot–Marschak (BDM) mechanism (Becker et al., 1964). The BDM is incentive-compatible because truthful reporting weakly dominates any other strategy when the drawn offer is binding and the respondent cannot influence its distribution (Berry et al., 2020). Enumerators framed the exercise as a real purchase, and respondents who accepted the drawn price paid in cash and received an EPC at the end of the visit.¹⁰ Appendix A reproduces the full enumerator script, comprehension prompts, and a practice jelly- or soap-purchase round that preceded the EPC elicitation.

The price ladder is country-specific. In Ethiopia and Kenya, respondents face 11 options at 0, 10, 20, . . . , 100 percent of the country-specific market price. In Uganda, respondents face 10 options at 0 UGX and at 30,000 UGX increments from 30,000 up to 270,000 UGX (the full market price). These tiers correspond to 0 and roughly 11.1, 22.2, . . . , 100 percent of market. The reference market price P_c^{market} for country c is the prevailing retail price for the EPC model used in the study at the time of enumeration in 2023. Across the three countries this corresponded to about USD 60–75. After the respondent reported the highest price tier she was willing to pay, she drew the binding offer herself from an opaque bag containing identical tokens, one per price tier on the country-specific ladder.¹¹

¹⁰Respondents who accepted the drawn price but did not have the cash on hand at the time of the visit could contact the field team within a one-month window after the visit to complete the purchase. In practice, on-the-spot cash payment was the norm in all three countries, and buyers received the physical EPC at the household visit. For the few delayed purchases, buyers called the original enumerator in Kenya and Uganda and the research team in Ethiopia. The EPC was then delivered to the household in Ethiopia and Kenya, while Uganda buyers collected it from the research team at Makerere University. Appendix A (§A.4) documents the per-country logistics.

¹¹We had respondents physically draw the offer rather than have the CAPI instrument generate it because of concern that respondents would not believe a computer-generated price was truly random. This parallels

The BDM module produces two distinct WTP measures, which we use for different analyses. First, the respondent reports the highest tier on the ladder she would accept. We record this as the ladder maximum share

$$M_i^{\text{share}} = \frac{M_i}{P_{c(i)}^{\text{market}}} \in [0, 1], \quad (1)$$

where i indexes the respondent (household), $c(i) \in \{\text{Ethiopia, Kenya, Uganda}\}$ is the country, M_i is the stated ladder maximum, and $P_{c(i)}^{\text{market}}$ is the reference market price in that country. M_i^{share} is hypothetical (the offer at M_i is not binding) and is defined for the full sample. Second, the respondent draws a binding price P_i from the bag. Under the BDM protocol the transaction is mechanical. The respondent purchases the EPC at P_i if and only if $P_i \leq M_i$, and otherwise walks away. This commitment rule is what delivers incentive compatibility, and enumerators explained it to respondents before the draw. We record the realized purchase as the binding choice

$$\text{buy}_i = \mathbf{1}\{\text{respondent accepts the binding offer at } P_i\} \in \{0, 1\}, \quad (2)$$

together with the realized price share $P_i^{\text{share}} = P_i / P_{c(i)}^{\text{market}}$. buy_i is revealed-preference (the transaction is binding) and is defined for households who completed the binding offer. In Ethiopia, enumerators did not administer the binding offer to the 22.75% of respondents who set $M_i = 0$.¹² We normalize both measures by $P_{c(i)}^{\text{market}}$ to preserve cross-country comparability of the demand schedule.

We use the realized BDM offer as a built-in validity check. Because the offer is drawn physically by the respondent from an opaque bag with identical tokens, the realized price should

the scratchcard implementation in [Burchardi, de Quidt, Gulesci, Lerva, and Tripodi \(2021\)](#), who motivate their physical randomizer with the same incentive-compatibility concern. If participants believe their stated WTP can influence the price, the mechanism's truthfulness property is lost.

¹²Ethiopia enumerators did not allow respondents who set $M_i = 0$ to draw a binding price. This affected 114 respondents (22.75% of the Ethiopia sample). Kenya and Uganda participation rates are approximately 99%. By construction the gated respondents are the lowest-WTP segment, so per-protocol estimates of buy_i overstate Ethiopia demand relative to the intent-to-treat statistic.

be orthogonal to household covariates within country. Figure B1 shows that the marginal distribution of realized draws is non-uniform within country. A chi-square goodness-of-fit test against discrete uniformity rejects in each country at the 1% level. The rejection is driven by a high-price skew in Kenya (over-draws at 80–100% of market price) and milder lumpiness in Ethiopia and Uganda. All three countries under-draw the zero tier.

Incentive-compatible BDM elicitation requires two conditions. The ladder must offer full support on prices, and respondents must not be able to manipulate the draw in a way correlated with their stated WTP (Becker et al., 1964; Burchardi et al., 2021). Uniformity of the marginal distribution is not required. We test the orthogonality condition directly by regressing the draw on the pre-treatment household covariate vector V_i . Table B1 regresses the drawn price share $P_i^{\text{share}} = P_i / P_{c(i)}^{\text{market}}$ on V_i , pooled with country fixed effects and separately by country. Inference uses cluster-robust standard errors and the wild-cluster bootstrap (Webb weights, 9,999 replications, via `boottest` (Roodman et al., 2019)).

The pooled wild-cluster bootstrap p -value is 0.72. By-country values are 0.76 (Ethiopia), 0.24 (Kenya), and 0.28 (Uganda). We cannot reject orthogonality at the 10% level in any column. Cluster-robust F -tests reject in the by-country columns. However, with 13–26 clusters per country and 12–13 covariates, the cluster-robust estimator is known to over-reject (Cameron et al., 2008; MacKinnon & Webb, 2017, 2018). We therefore base inference on the wild-cluster bootstrap.¹³

4.2 Diagnosing overstatement in the stated ladder

The BDM module generates a within-subject test of the stated ladder. For households whose drawn price falls at or below their stated maximum ($P_i \leq M_i$), the commitment rule requires purchase. A refusal directly reveals that the stated maximum overstated true WTP. We call

¹³We ask the `know_EPC` question (prior familiarity with electric pressure cookers) at the start of the cooking-technology module, before any random assignment to the information treatment. We therefore treat it as a pre-treatment covariate in all balance and outcome regressions.

these households the at-risk sample and define

$$\text{overstate}_i = \mathbf{1}\{P_i \leq M_i \text{ and } \text{buy}_i = 0\}, \quad (3)$$

on the at-risk sample. The reverse inconsistency, buying at a drawn price above the stated maximum, identifies understatement. It occurs in 2 households across the full sample and we set it aside.

Two diagnostic regressions discipline the interpretation of overstatement. The first estimates the refusal gradient within the at-risk sample:

$$\text{overstate}_i = \alpha + \delta P_i^{\text{share}} + \lambda M_i^{\text{share}} + \gamma_{c(i)} + \varepsilon_i, \quad (4)$$

by OLS (linear probability model). Because the at-risk sample conditions on $P_i \leq M_i$, the unconditional refusal-price slope confounds behavior with sample composition. Higher draws mechanically survive only for high- M households, so a rising raw slope can emerge even if stated maxima were pure noise. We address this in two ways. First, we verify the confound directly with a permutation test that reshuffles M_i across households within country, rebuilds the at-risk sample, and re-estimates the raw slope. The resulting null distribution quantifies how much gradient selection alone produces. Second, equation (4) conditions on M_i^{share} . The draw is random, so holding M_i fixed, δ isolates the behavioral price response, and λ tests whether the stated maximum carries information, with $\lambda = 0$ under uninformative ladders and $\lambda < 0$ if higher stated WTP marks genuinely higher valuation. If respondents strategically inflated the ladder to raise the chance of drawing a low binding price, stated maxima would instead pile up at the top tier. Section 5.2 reports the results.

The second diagnostic asks whether overstatement is predictable at the household level:

$$\Pr(\text{overstate}_i = 1 \mid W_i) = \Phi(W_i' \theta + \gamma_{c(i)}), \quad (5)$$

a probit on the at-risk sample, where W_i collects respondent education, enumerator-assessed BDM comprehension, monthly expenditure, and the household’s stated ladder maximum from the practice round. The practice round has each respondent complete a full ladder and binding draw for a small non-energy good (jelly or soap; Appendix A) before the EPC module. The practice-round ladder share serves as a placebo regressor. It carries no information about EPC valuation, so any predictive power it has for EPC overstatement isolates a generic response style rather than an EPC-specific consideration. If (5) classified overstaters accurately, one could correct each household’s stated WTP directly. Section 5.3 shows the classifier is too weak and too country-driven to support that correction, which motivates the bounds we construct next.

4.3 Bounding the demand curve

We construct three demand curves per country that bracket true demand. Each curve reports, at every price tier ps on the country ladder, the share of households willing to buy at that price.

Curve A (stated ladder). The share of households with $M_i^{\text{share}} \geq ps$, computed on the full sample. If households overstate, curve A overstates demand at every price and understates the subsidy required for any adoption target.

Curve B (interior estimate). A precision-weighted blend in which each household contributes its ladder cells with weight $1 - \hat{p}_i$ and, if it completed a draw, its binding outcome with weight $\hat{p}_i$, where $\hat{p}_i$ is the predicted overstatement probability from (5). Curve B is an interior estimate, not a corrected demand curve. It inherits the classifier’s weakness, and we report it to show that a plausible interior construction lands between the stated and binding curves rather than at either boundary.

Curve C (binding only). The share of households accepting the binding offer among those who drew price ps , computed on the per-protocol BDM sample. Curve C is revealed preference, but each tier rests on the subsample that happened to draw it, and the per-protocol gating in Ethiopia (§4.1) tilts its sample toward the higher-WTP segment.

From each curve we compute the subsidy consistent with 25%, 50%, and 75% adoption by linear interpolation between adjacent price tiers. The pair (A, C) bounds the subsidy requirement. The distance between the two measures what stated-preference data alone cannot pin down. As an independent coherence check, we also recompute curve A after capping each observed overstater’s ladder maximum at 10 percentage points below the household’s drawn price. Section 5.4 shows this alternative construction lands inside the (A, C) interval at every adoption target.

4.4 Information treatment

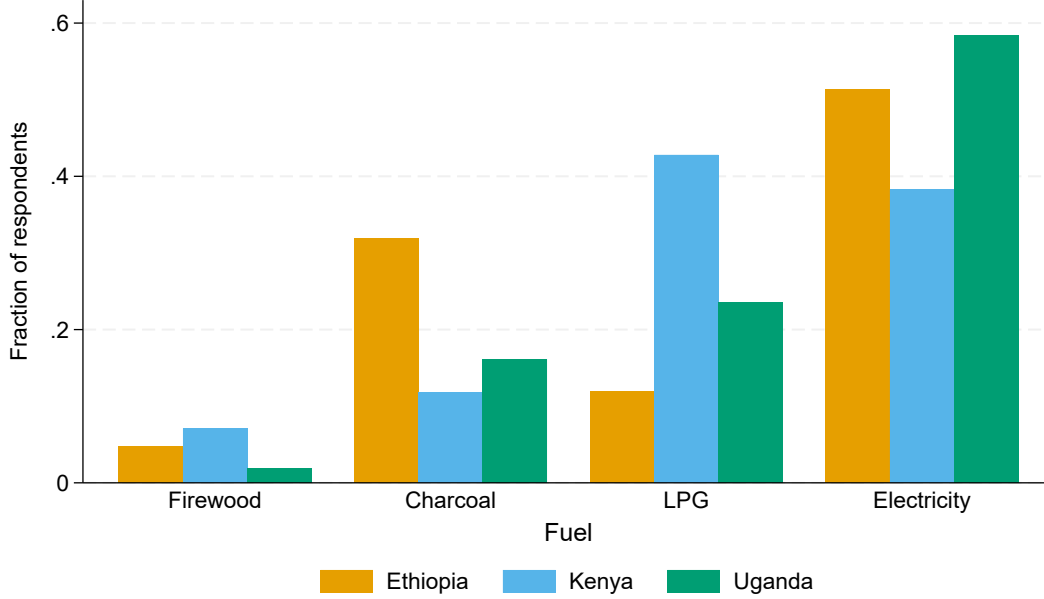
In prior fieldwork in East Africa, researchers have documented a recurring perception that electric cooking is unaffordable for households outside the top of the income distribution. Kersey et al. (2024) record a Kampala informal-settlement restaurateur summarizing the view as “[cooking with] electricity is for the rich.” Across 25 informal communities, they identify “fear that [electric cooking appliances] would dramatically increase electricity consumption and hence costs” as a recurring misconception. Leary and Onjala (2022) document the same perception in Kenya. Our own pre-treatment survey data confirm the pattern. In our sample, 47.3% of respondents identify electricity as the most expensive fuel for cooking a single staple dish, ahead of LPG (27.0%), charcoal (20.4%), and firewood (5.1%) (Figure 2; $N = 1,306$).¹⁴

This perception may be accurate for conventional resistive electric cooking (coil stoves, hot plates), which consume substantial power per meal (Leary & Onjala, 2022). However, the EPC’s high thermal efficiency and short cook times materially change the per-meal cost calculation. Using MECS-program cross-country data, Leary and Onjala (2022) report that EPC operation costs roughly one-third of LPG and less than 20% of charcoal. Country-specific cost-per-meal figures in the MECS Kitchen Laboratory report for Ethiopia and the Kenya and Uganda eCookbooks place EPC operation costs below charcoal and LPG across our three settings

¹⁴The exact wording was: “Consider cooking one dish of [local staple food]. Of the following fuels, please tell me which you think is the most expensive, i.e. the fuel required to cook one staple dish has the highest cost? The fuels are: firewood, charcoal, LPG, electricity.” We asked the question at the start of the cooking-technology module, before any random assignment to the information treatment.

(Lakew Tesema et al., 2020; Leary et al., 2019; Naluwagga et al., 2022). We therefore designed the information treatment to correct this perception with EPC-specific cost-per-meal evidence.

Figure 2: Perceived costliest cooking fuel, by country



Notes: Share of respondents identifying each fuel as the most expensive cooking fuel they could use, by country. Asked at the start of the cooking-technology module, before any random assignment to the information treatment.

We randomly assigned half of households to receive an information treatment immediately before the EPC BDM. Assignment used simple randomization at the household level, implemented via a random-number draw inside the CAPI instrument. Wild-cluster bootstrap joint F -tests cannot reject orthogonality of the full covariate vector to treatment assignment in any column (Table D1; pooled $p = 0.15$, by-country $p \geq 0.70$). However, five of thirteen baseline covariates, primarily fuel-stack indicators and outage exposure, are individually imbalanced in the pooled column at the 10% level under wild-cluster bootstrap. We therefore include these covariates as controls in robustness columns of Table 3 and Appendix Table D3.

We tailored the treatment content to local fuels, foods, and prices in each country. In Ethiopia, enumerators showed treated respondents results from the MECS Kitchen Laboratory report for Ethiopia (Lakew Tesema et al., 2020). In Kenya and Uganda, enumerators showed treated respondents the corresponding national eCookbooks (Leary et al., 2019; Naluwagga

et al., 2022). Each source reports the cooking cost per standard meal or recipe across cooking technologies (EPC, charcoal, kerosene, LPG, and where relevant firewood), using a menu of national staple dishes. The operative message was identical in direction in all three countries. Cooking the same menu on an EPC costs less per meal than on any of the available alternatives, including charcoal. Enumerators walked each treated respondent through a one-page summary table before initiating the BDM ladder.

The estimand is the intent-to-treat (ITT) effect of being offered the information on M_i^{share} , denoted τ in equation (8) below. τ recovers the effect of being offered the information, which is the policy-relevant counterfactual for information campaigns that anchor many clean-cooking promotion programs in low- and middle-income countries.

4.5 Estimating equations

Beyond the diagnostic and bounding constructions of §4.2–4.3, we estimate three specifications in the body of the paper, plus a built-in BDM-price-exogeneity check estimated in Appendix B. Each specification uses the WTP measure best matched to its question. The revealed-preference demand schedule uses the binding choice buy_i , and the correlates and information-treatment analyses use the ladder maximum M_i^{share} , which supplies a continuous outcome on the full sample.

The first specification estimates the demand schedule from the binding BDM choice. Let P_i^{share} denote the realized BDM price share. For each country c we estimate

$$\Pr(\text{buy}_i = 1 \mid P_i^{\text{share}}, c(i) = c) = \Phi(\alpha_c + \beta_c P_i^{\text{share}}), \quad (6)$$

by maximum-likelihood probit, where $\Phi(\cdot)$ is the standard normal CDF. The probit form is equivalent to a structural model with Gaussian latent willingness to pay $M_i^* \mid c \sim N(\mu_c, \sigma_c^2)$, where $\alpha_c = \mu_c/\sigma_c$ and $\beta_c = -1/\sigma_c$. The choice probability $\Phi(\alpha_c + \beta_c P_i^{\text{share}})$ then equals $\Pr(M_i^* \geq P_i^{\text{share}} \mid c)$, the survival function of latent WTP at the offered price. The price at 50% adoption, $P^{50,c} = -\alpha_c/\beta_c$, recovers the country- c mean latent WTP μ_c and is the headline

subsidy-for-adoption target reported in Figure 3 and Appendix Table D5. Because the BDM draw is orthogonal to household covariates within country (verified in §4.1 and Table B1), β_c identifies the country- c price slope of acceptance without further controls.

The second specification characterizes the correlates of WTP using the continuous ladder measure on the full sample:

$$M_i^{\text{share}} = \alpha + X_i' \beta + \gamma_{c(i)} + \varepsilon_i, \quad (7)$$

where X_i is a vector of household-level covariates and $\gamma_{c(i)}$ are country fixed effects. We estimate (7) both pooled with country fixed effects and separately by country to isolate within-country variation.

The third specification recovers the ITT effect of information:

$$M_i^{\text{share}} = \alpha + \tau D_i + C_i' \beta + \gamma_{c(i)} + \varepsilon_i, \quad (8)$$

where $D_i \in \{0, 1\}$ is the information treatment indicator, τ is the ITT, and C_i is a vector of household-level covariates selected to absorb the baseline imbalances flagged in §4.4. The random assignment of D_i identifies τ without further assumptions. C_i enters to tighten inference and absorb the flagged imbalances. As a secondary specification we re-estimate (8) replacing M_i^{share} with the binding choice buy_i , conditioning on the realized draw P_i^{share} . This isolates the information effect on revealed demand at the binding offer.

4.6 Inference

We cluster standard errors at the unit of sampling (Abadie, Athey, Imbens, & Wooldridge, 2023).¹⁵ The clusters are the district in Ethiopia (13 clusters), the ward in Kenya (26 clusters), and the village in Uganda (24 clusters). The three by-country samples pool to 63 clusters in the cross-country specifications. Appendix Table D2 reports the cluster size distribution by country.

¹⁵Ethiopia is the exception: we sampled households from EEU customer books, but book identifiers were not retained in the data, so we cluster at the district level there.

Cluster-robust variance estimates use Stata’s default $G/(G - 1)$ small-sample correction. Even with this correction, the conventional cluster-robust estimator over-rejects in finite samples when $G < 30$ (Cameron et al., 2008), which binds for each by-country specification.

We therefore report wild-cluster bootstrap p -values alongside the cluster-robust p -values for every hypothesis test in the paper, including the refusal-gradient and overstatement regressions of §4.2. The bootstrap uses Webb six-point weights and 9,999 replications (MacKinnon & Webb, 2018; Roodman et al., 2019). It applies to both single-coefficient t -tests (the ITT τ in (8)) and joint F -tests (the correlates test in (7) and the BDM-price exogeneity test in Appendix B). Webb weights are the recommended choice when the smallest cluster count falls below 30 (Webb, 2014), as in our Ethiopia sample. Where the cluster-robust and bootstrap p -values diverge, we treat the bootstrap as the preferred inference, following the recommendation of MacKinnon and Webb (2018); Roodman et al. (2019).

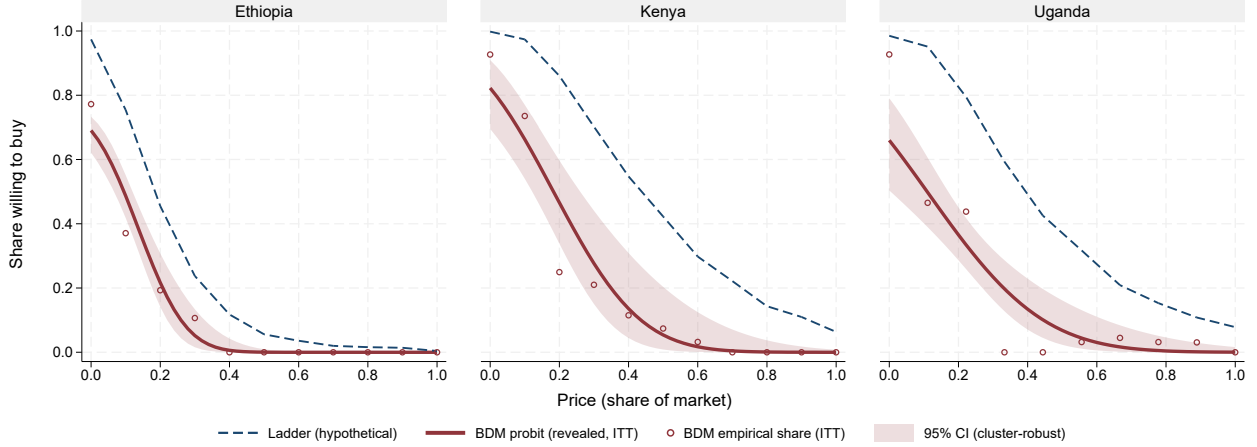
We report Romano-Wolf step-down p -values (Romano & Wolf, 2005; Clarke, Romano, & Wolf, 2020) alongside the per-equation bootstrap p -values to control the family-wise error rate (FWER) across three families of hypotheses, selected by substantive interpretability rather than statistical significance.¹⁶ The families are (i) the by-country ITT triplet from (8) estimated separately in Ethiopia, Kenya, and Uganda (3 tests); (ii) the heterogeneity family for the ITT, comprising five interactions of D_i with respondent gender, household-head gender, main fuel charcoal, main fuel LPG, and baseline EPC familiarity (5 tests); and (iii) the correlates family, the four coefficients from (7) we interpret substantively in the text: household expenditure (inverse-hyperbolic-sine of USD), household-head gender, baseline EPC familiarity, and outage hours per week (4 tests). The pooled ITT τ from (8) is a single hypothesis, so FWER correction over that singleton is moot. We implement the Romano-Wolf step with the `rwolf2` Stata package using cluster bootstrap at the neighborhood level with 9,999 replications and a fixed seed.

¹⁶Romano-Wolf is mechanically valid for any fixed family of hypotheses, but the FWER guarantee is most credible when the family is chosen before the analyst sees the regression output (Burlig, 2018). We disclose the post-hoc family construction and report all coefficients within each family.

5 Results and Discussion

5.1 Stated and revealed demand curves

Figure 3: EPC demand is sharply downward sloping, and ladder elicitation overstates demand



Notes: Per-country demand curves for EPCs. The thick maroon line is the headline BDM-revealed probit smoother (binding choice at the randomly drawn price); the shaded region is the pointwise 95% confidence band for the probit fit (delta method on the latent index, standard errors clustered at the neighborhood level, t_{G-1} critical values); maroon circles are the within-tier empirical shares; the navy dotted line is the hypothetical ladder demand curve. The BDM curve, band, and empirical shares are intention-to-treat: the probit is fit to BDM participants and scaled by the participation share π_c (387/501 in Ethiopia, 536/537 in Kenya, 264/268 in Uganda), coding households never offered a draw as non-buyers at every positive price (π_c treated as known). Per-country ITT sample sizes: Ethiopia $N = 501$; Kenya $N = 537$; Uganda $N = 268$.

Demand for EPCs is sharply downward sloping, and the implied subsidies required for even modest adoption are deep. The BDM-revealed probit demand curve in Figure 3 (tabulated in Appendix Table D5) traces the share of households accepting the binding offer as a function of the realized price. Reaching 50% country-wide adoption requires a price of 9.5% of market in Ethiopia (95% CI: 7.0–12.1), 18.3% in Kenya (95% CI: 12.0–24.6), and 11.0% in Uganda (95% CI: 3.7–18.2), corresponding to subsidies of 90.5%, 81.7%, and 89.0%, respectively.¹⁷ At the full market price, predicted take-up is below 5% in all three countries.

¹⁷These are intention-to-treat estimates. The demand curve is $\pi_c \Phi(\hat{\alpha}_c + \hat{\beta}_c p)$, where π_c is the BDM participation share (387/501 in Ethiopia, 536/537 in Kenya, 264/268 in Uganda). Non-participants, all of whom stated zero WTP in Ethiopia, are coded as non-buyers at every positive price. The price at 50% adoption solves $\Phi(\hat{\alpha}_c + \hat{\beta}_c p) = 0.5/\pi_c$. Confidence intervals are delta-method intervals with standard errors clustered at the neighborhood level, treating π_c as known. The per-protocol estimates, conditioning on participation, are 13.6% (subsidy 86.4%) in Ethiopia, 18.4% (81.6%) in Kenya, and 11.4% (88.6%) in Uganda.

The ladder elicitation overstates demand relative to the binding choice in every country, though the magnitude varies sharply. The hypothetical ladder curve implies subsidies of 81.5%, 56.2%, and 60.5% for 50% adoption in Ethiopia, Kenya, and Uganda. These are below the BDM-revealed subsidies of 90.5%, 81.7%, and 89.0% by 9.0 percentage points in Ethiopia, 25.5 in Kenya, and 28.5 in Uganda. Both curves are estimated on the full sample. The stated ladder includes the Ethiopian zero-WTP households directly, and the ITT binding curve counts them as non-buyers, so the comparison holds the population fixed.

5.2 Stated versus revealed demand: the refusal gradient

The ladder–BDM gap is concentrated among households who said they would buy at the binding price but ultimately did not. Among the at-risk sample (households whose drawn BDM price fell at or below their stated ladder WTP, in which a binding refusal reveals overestimation), 62.1% refused the binding offer (211 of 340). The rate is substantial in all three countries but varies: 44.0% in Ethiopia (33 of 75), 72.3% in Kenya (115 of 159), and 59.4% in Uganda (63 of 106).¹⁸

Refusal rises steeply with the drawn price. Within the at-risk sample, the refusal rate climbs from 3.9% at the lowest positive price tiers to 100% at the top of the drawn-price distribution. The raw pooled slope is 0.0105 per percentage point of market price (wild-cluster bootstrap $p < 0.001$). However, the raw slope is not by itself evidence about the stated ladder. The permutation test of §4.2 reshuffles stated maxima within country and rebuilds the at-risk sample 2,000 times. It shows that selection on $P_i \leq M_i$ combined with downward-sloping demand generates a slope of 0.0127 (permutation sd 0.0006) even when stated maxima carry no information. The observed slope of 0.0105 falls below all 2,000 selection-only draws. The shortfall is itself informative. Under uninformative ladders, high draws would survive into the at-risk sample at random, and refusal would climb at the mechanical rate. Instead, high draws

¹⁸The reverse inconsistency, buying despite a drawn price above ladder WTP, occurs in only 2 households across the full sample. The at-risk denominator uses the per-protocol BDM sample (Section 4.1). The 114 Ethiopia respondents who set $M_i = 0$ and were not offered a draw are excluded by construction because they did not draw a binding price (119 non-participants pooled, including 1 in Kenya and 4 in Uganda; §4.1).

survive only for households whose stated maxima mark genuinely higher valuations, and those households follow through more often than the selection-only benchmark predicts.

The conditional estimates carry the diagnostic content (Appendix Table D7). Estimating equation (4) with the stated maximum included, the price slope is 0.0125 per percentage point of market price (cluster-robust SE 0.0011; wild-cluster bootstrap $p < 0.001$), and the stated maximum enters at -0.0037 per percentage point (SE 0.0010): at any given drawn price, households who stated higher WTP are more likely to follow through. Because the draw is random, the conditional slope is free of the selection confound, and the negative coefficient on the stated maximum rejects the uninformative-ladder null directly. Country-by-M-tier fixed effects in place of the linear control give a nearly identical price slope (0.0126).

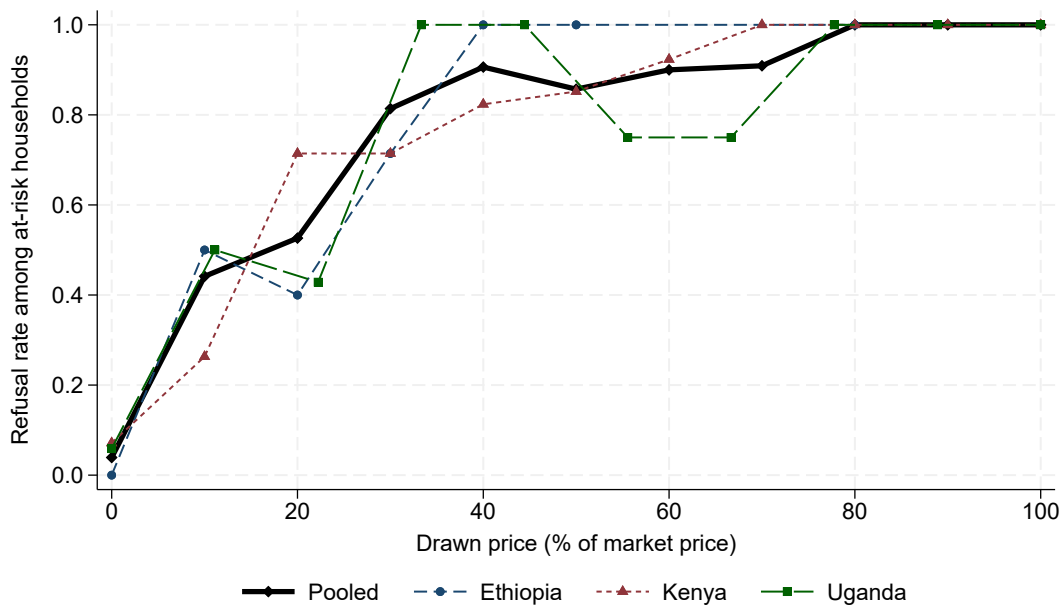


Figure 4: Refusal of the binding offer rises steeply with the drawn price

Notes: Each point is the mean refusal rate among at-risk households (drawn price at or below stated ladder WTP; $N = 340$) at a drawn-price tier, expressed as a percent of the country market price. The pooled series rounds Uganda's ninths-based tiers to the nearest 10-percentage-point tier. Tier-level cell sizes are small at the extremes of the drawn-price distribution.

The conditional estimates discriminate between models of the stated ladder. Households refuse when the draw lands far above what they are truly willing to pay, and comply

when it lands well below. The steep conditional price slope shows that stated maximums sit systematically above the prices households accept.

The gradient also constrains strategic explanations. A respondent gaming the mechanism would inflate the stated maximum to raise the chance of drawing a low binding price, and so would state the top tier. Almost no one does. 0.4% of Ethiopian respondents (2 of 501), 6.3% of Kenyan respondents (34 of 537), and 7.8% of Ugandan respondents (21 of 268) state a ladder maximum equal to the full market price. The joint pattern, an informative stated maximum, a steep conditional refusal gradient, and no mass at the ladder top, is inconsistent with both pure hypothetical bias (uninformative stated responses) and crude gaming. We cannot rule out subtler strategic responses of the kind formalized by [Carson et al. \(2014\)](#), but the data are most consistent with stated WTP that is systematically inflated yet informative about the ordering of households.

A same-day cash shortfall cannot explain the refusals. The binding offer included a callback option: respondents who wanted the EPC but lacked cash on hand at the visit could complete the purchase up to one month later, and we code $\text{buy}_i = 1$ for these completions. A binding “no” therefore reflects a household that neither paid on the spot nor returned within the callback window, so a transitory liquidity shock alone cannot account for the gap.

Enumerator-reported BDM understanding is high in all three countries (Appendix Table D6). The top two categories (“Very good” or “Good”) cover 89.8% of Ethiopian, 96.1% of Kenyan, and 97.8% of Ugandan respondents. Restricting to high-understanding respondents barely moves either curve (Appendix Figure D2). The ladder P^{50} shifts by at most 1.1 percentage points across the three countries, and the BDM probit P^{50} by at most 0.5 percentage points. The ladder–BDM gap is therefore not a comprehension artifact.

One mechanism remains consistent with the pattern: hypothetical bias amplified by a decision-context wedge between the ladder and the binding offer. The ladder asks respondents to imagine paying a sequence of prices. The binding offer requires them to hand over cash for a bulky, unfamiliar durable they have likely never owned. The cash-loss is salient only

at the binding moment, and the size and novelty of the EPC could trigger reconsideration that the abstract ladder exercise does not surface. Smaller-good BDM experiments report substantially smaller ladder-binding gaps (Berry et al., 2020; Berkouwer & Dean, 2022), consistent with the decision-context interpretation. Experiments that hold economic incentives fixed across question formats find little independent role for behavioral framing (Vossler & Zawojkska, 2020), which also points to the decision context rather than the ladder format itself. However, we cannot rule out a contribution from imperfect BDM comprehension. Even with the practice jelly round, the binding mechanism is unfamiliar to most respondents, and a residual comprehension failure would generate similar refusals. Such game-form misconceptions inflate valuation gaps in stated-preference settings even under incentive-compatible mechanisms (Vossler, Bergeron, Doyon, & Rondeau, 2023).

5.3 Who overstates?

Overstatement is weakly predictable, and the predictors point to a generic response style. The probit in equation (5), estimated on the 340 at-risk households, returns three average marginal effects of interest (Appendix Table D8). Each additional year of respondent education reduces the probability of overstating by 3.1 percentage points ($p = 0.045$). Each percentage point of the practice-round (jelly) ladder share raises it by 0.3 percentage points ($p = 0.009$): households who state high WTP for a trivial non-energy good are also the ones who overstate for the EPC, which is the signature of a survey response style rather than an EPC-specific consideration. Enumerator-assessed BDM comprehension does not predict overstatement (AME 0.067, $p = 0.129$), so comprehension failures do not drive the gap.

The classifier is too weak to support a household-level correction. The probit’s pseudo- R^2 is 0.081, and it classifies 66.5% of at-risk households correctly against a 62.1% base rate – a 4.4 percentage-point improvement over predicting “overstates” for everyone. Extrapolated to the full sample, the fitted model flags 97% of Kenyan households as predicted overstaters, so a trim-the-overstaters correction would rest on a handful of

unflagged households in Kenya. We therefore do not correct stated WTP household by household. We bound the demand curve instead (§5.4).

The remaining question is who states high WTP in the first place. Appendix C reports and discusses the full correlates analysis (equation (7); Appendix Table C1). We summarize it here. Monthly expenditure is the strongest single correlate of WTP. Doubling monthly expenditure is associated with a 2.86 percentage-point increase in WTP (s.e. 0.63) in the pooled specification with country fixed effects, an association that survives Romano-Wolf correction at the 1% level and is consistent with the credit-constraint mechanism of Berkouwer and Dean (2022). Baseline fuel use moves WTP in the direction opportunity cost predicts. LPG-main households state WTP 4.90 pp higher (s.e. 2.19, $p < 0.05$) and charcoal-main households 4.40 pp lower (s.e. 1.68, $p < 0.01$) than the within-country reference category. Prior EPC familiarity is positive but suggestive only (RW $p = 0.088$). Household demographics, decision-rights proxies, elicited time and variance preferences, and within-country outage exposure are all null. The expenditure and fuel associations are robust to the four ladder-consistency adjustments of Appendix Table D9; only grid access attenuates, consistent with off-grid households being over-represented among ladder-overestimators (§5.2).

5.4 Bounded demand and subsidy thresholds

Stated-preference data understate the subsidy required for adoption targets by 9 to 29 percentage points. Table 2 reports the subsidy consistent with 25%, 50%, and 75% adoption under each demand curve of §4.3. At the 50% target, the stated ladder (curve A) implies subsidies of 81.5% in Ethiopia, 56.2% in Kenya, and 60.5% in Uganda. The ITT probit fitted to binding choices (§5.1) implies 90.5%, 81.7%, and 89.0%, and the binding-only interpolation (curve C, per-protocol) gives 90.4%, 85.1%, and 89.5%. A planner who set Kenya’s subsidy from the stated ladder would fund 56% of the EPC price and reach half of urban households only if stated responses were truthful. The revealed data say the same target requires 82%. The gap between the probit and the stated ladder is smallest in Ethiopia (9.0 percentage points), where the stated ladder is already steep, and largest in Kenya (25.5) and Uganda (28.5).

Appendix Table D13 reports the ITT and per-protocol probit estimates side by side, with BDM participation shares and delta-method confidence intervals. The two differ materially only in Ethiopia, where the 114 gated ladder-zero households enter the ITT denominator.

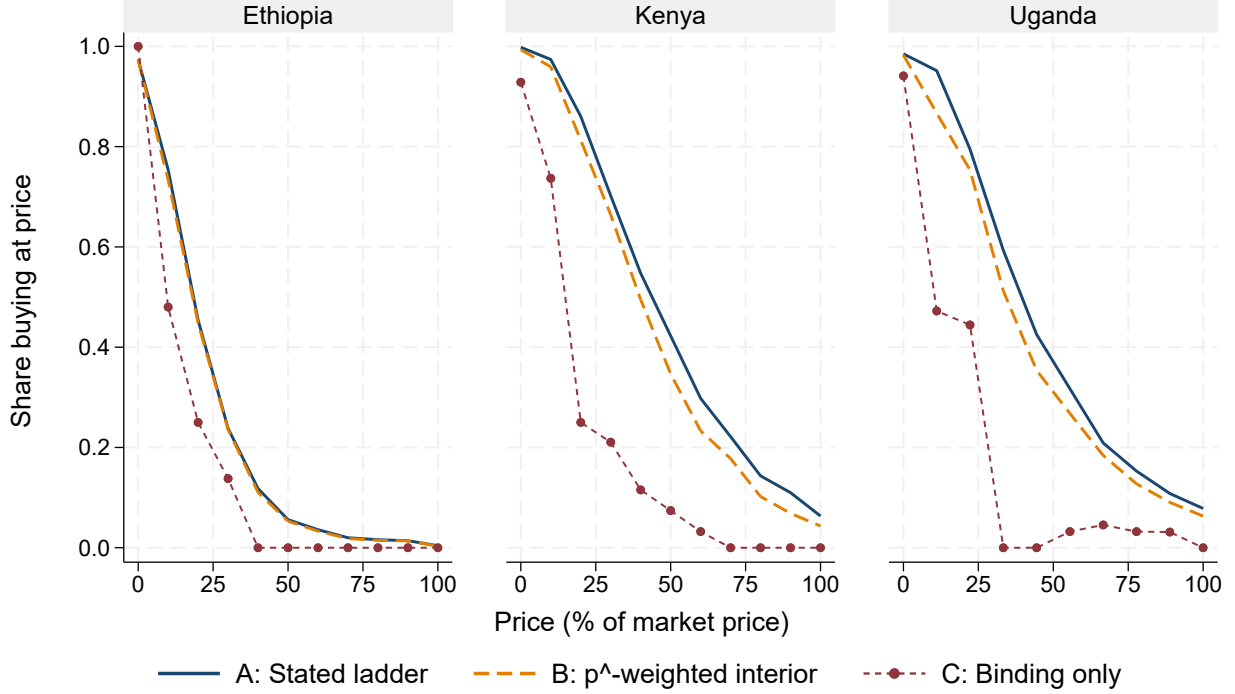


Figure 5: Stated and binding demand curves bracket the interior estimate in every country
Notes: Each panel plots the share of households buying at each price, expressed as a percent of the country market price. Curve A uses the full stated ladder; curve B weights every household's ladder cells by $1 - \hat{p}_i$ and its binding outcome by $\hat{p}_i$, where $\hat{p}_i$ is the fitted overstatement probability from equation (5); curve C uses binding BDM outcomes only (per-protocol sample), so its tier-level cells are small. Table 2 reports the implied subsidy thresholds.

The bounds are informative. The ordering $A \leq B \leq C$ holds at every adoption target in every country. The two interior estimates built from different inputs – the $\hat{p}$ -weighted curve B (predicted overstatement) and the cap-at-draw construction (observed overstatement) – land inside the (A, C) interval at all nine country-by-target cells. In Ethiopia, the cap construction and curve B agree almost exactly at the 50% target (82.2 versus 81.8). In Kenya the two disagree more (70.3 versus 60.3), which is the honest signature of a weak classifier. Both remain well inside the 56.2–85.1 bracket. For subsidy design, the

Table 2: Subsidy required for 25%, 50%, and 75% adoption, by demand curve

Adoption target	Ethiopia			Kenya			Uganda		
	25%	50%	75%	25%	50%	75%	25%	50%	75%
A: Stated ladder (full sample)	70.6	81.5	89.9	33.7	56.2	73.0	37.5	60.5	75.3
B: $\hat{p}$ -weighted interior	70.7	81.8	90.7	41.5	60.3	75.8	42.1	65.8	77.6
C: Binding only	80.0	90.4	95.2	80.0	85.1	90.7	72.9	89.5	95.5
Cap at draw – 10pp (check)	74.6	82.2	90.7	45.9	70.3	77.6	58.1	76.2	89.5

Notes: Each cell reports the subsidy (percent of market price) at which the indicated demand curve reaches the column’s adoption share, by linear interpolation between adjacent price tiers. Curve A uses the full stated ladder; curve B weights every household’s ladder cells by $1 - \hat{p}_i$ and its binding outcome by $\hat{p}_i$, where $\hat{p}_i$ is the predicted overstatement probability from equation (5); curve C uses binding BDM outcomes only (per-protocol sample). The final row recomputes curve A after capping each observed overstater’s ladder maximum at 10 percentage points below the household’s drawn price – an independent construction that uses actual, not predicted, overstatement. Curve B inherits the classifier’s weakness (§5.3) and is reported as an interior estimate, not a corrected demand curve.

operational reading is the bracket itself: stated data give the floor, binding data the ceiling, and any defensible correction lands in between.

Curve C carries two caveats that argue for treating it as a ceiling rather than the truth. Each price tier of curve C rests on the subsample of households that drew that tier, so tier-level cells are small, and the Ethiopia per-protocol sample excludes the 114 gated ladder-zero households, tilting Ethiopian revealed demand upward (§4.1). Neither caveat rescues the stated ladder: stated-preference targeting is adequate only if the true requirement sits at the bottom of the bracket, which would require curve C to overstate the subsidy by its full 29-point width in Kenya and Uganda – far beyond what small tier cells or the Ethiopia gating can produce.

5.5 Mechanism: good economics, binding liquidity

Refusals are not explained by poor economics: the EPC pays for itself within months. Using country tariffs and fuel prices prevailing at enumeration, the median household’s fuel-cost savings repay the EPC purchase price in 2.1 to 4.8 months at the drawn BDM price and in 4.7 to 7.8 months at the full market price, across the three countries. The median break-even point is 253 to 495 meals, against roughly 3,650 meals over a conservative appliance life. Only 6.5% of households have non-positive predicted savings (Appendix Table D14).

Refusal tracks the sticker price, not the payback calculation. Conditional on the drawn price, the log of the household-specific payback period does not predict refusal (average marginal effect 0.003, $p = 0.88$). Households that overstate do have longer paybacks than compliers (medians 1.3–2.9 versus 0.2–0.7 months at the drawn price), but once the price itself is held fixed, variation in the return does not move the refusal margin. If households were trading off returns against costs, the return should matter. It does not.

Two facts point to liquidity constraints combined with non-salient savings. First, the EPC market price equals 47% to 67% of monthly household expenditure across the three countries. This is a lump sum comparable to what [Berkouwer and Dean \(2022\)](#) show Nairobi households cannot finance for a charcoal stove with a 295% annual return absent credit. Second, roughly 70% of respondents expect zero fuel-cost savings from an EPC, so the asset’s return is invisible to most buyers at the moment of purchase. A household that cannot smooth a lump sum equal to half a month’s budget, and does not believe the savings are real, rationally refuses even a deeply discounted offer. This reading implies the binding margin is financing, not information. It is a prediction the information experiment tests directly.

5.6 Effect of the information treatment

The information treatment does not shift willingness to pay. The null is the outcome the mechanism evidence predicts: if the binding constraint is financing rather than beliefs (§5.5), correcting cost misperceptions should not move demand. In the pooled sample ($N = 1,306$), the point estimate is -0.74 percentage points of market price once country fixed effects are included (Table 3, column 3; $SE = 1.34$). However, the controls-only specification yields a larger negative estimate of -2.21 pp (column 2; $SE = 1.40$), which shrinks toward zero under country fixed effects – Ethiopia has both the lowest mean WTP and the strongest covariate set, so cross-country variation pulls the pooled coefficient negative until the fixed effects absorb it. Cluster-robust standard errors at the neighborhood level (63 clusters; column 5) leave the column 3 inference unchanged.

Table 3: Effect of information treatment on WTP – pooled sample

	(1) Baseline	(2) +Controls	(3) +Country FE	(4) +Interaction	(5) Cluster SE
Info treatment	-0.840 (1.489)	-2.205 (1.396)	-0.744 (1.339)	-0.398 (1.816)	-0.744 (1.300)
Respondent female		0.440 (2.042)	-0.838 (1.922)	-0.846 (1.924)	-0.838 (2.009)
Household size		0.376 (0.316)	0.648** (0.305)	0.642** (0.306)	0.648** (0.321)
Main cooking fuel: charcoal		6.436*** (1.820)	-5.099*** (1.728)	-5.131*** (1.732)	-5.099*** (1.862)
Main cooking fuel: LPG		18.618*** (1.693)	6.107*** (2.190)	6.100*** (2.192)	6.107** (2.475)
Typical outage hrs/week		-0.250*** (0.046)	-0.013 (0.045)	-0.013 (0.045)	-0.013 (0.048)
Elec. most expensive				-0.131 (1.816)	
Info treatment $\times$ Elec. most expensive				-0.760 (2.643)	
Constant	33.086*** (1.012)	25.886*** (2.704)	15.634*** (2.641)	15.759*** (2.814)	15.634*** (3.037)
Observations	1306	1304	1304	1304	1304
R^2	0.000	0.136	0.227	0.227	0.227

Notes: OLS estimates of the information treatment on WTP (share of market price, 0–100). Controls in columns 2–5: respondent female, household size, main cooking fuel is charcoal, main cooking fuel is LPG (reference category: any other primary fuel), typical weekly outage hours (winsorized at country-specific p95). Country fixed effects in columns 3–5. Column 4 interacts treatment with the indicator that the respondent perceives electricity as the most expensive cooking fuel. Column 5 reports cluster-robust standard errors at the neighborhood level (63 clusters); columns 1–4 use heteroskedasticity-robust standard errors. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Any treatment effect should concentrate on respondents who entered the script viewing electricity as the most expensive cooking fuel. The script’s comparative operating-cost table is informative precisely for these respondents – those who already place electricity below charcoal or LPG on the cost ladder are receiving a message that confirms their priors and should not update. Column 4 of Table 3 tests this *ex ante* prediction by interacting the treatment with an indicator that the respondent named electricity as the most expensive cooking fuel at baseline (elicited before treatment assignment; Figure 2). However, the interaction is also null: the implied treatment effect for the elec-expensive group is -1.16pp (a main effect of -0.40pp plus an interaction of -0.76pp , SE on the interaction = 2.64), and we cannot reject equality with the non-elec-expensive group at any conventional level.

Country-by-country estimates change sign but remain statistically null. We estimate -1.29pp in Ethiopia (SE = 1.50), -1.67pp in Kenya (SE = 2.36), and $+3.11\text{pp}$ in Uganda (SE = 3.82). All 95% confidence intervals cover zero (Appendix Table D3, columns 1, 3, 5). Wild-cluster bootstrap *p*-values widen rather than tighten this conclusion (0.277 in Ethiopia, 0.510 in Kenya, 0.541 in Uganda), consistent with small cluster counts (13, 26, 24). A Romano-Wolf step-down adjustment across the three-country ITT family yields RW $p = 0.347$ for each of Ethiopia, Kenya, and Uganda, leaving the null conclusion intact. Figure 6 plots all six estimates with their 95% intervals: the zero line crosses every interval, and the Uganda interval is wide enough to admit double-digit effects in either direction.

We rule out positive average treatment effects larger than $+1.9\text{pp}$ of market price, and negative effects larger than -3.4pp , in the pooled sample (95% confidence interval around the column 3 estimate). That ceiling is small relative to what would close the gap between elicited WTP and the unsubsidized price: shifting the demand schedule so that 25% adopt at market price would require a WTP increase of 78.6pp in Ethiopia, 68.1pp in Kenya, and 70.6pp in Uganda (§5.1). Even the upper end of Uganda’s wider per-country interval ($+10\text{pp}$) covers only about a seventh of that gap.

Heterogeneity tests across expenditure, cooking-fuel use, perceived electricity expense, and gender are uniformly null, except for one marginal respondent-gender interaction that points away from the cooking-budget channel we anticipated. Appendix Table D10 interacts the treatment with five observable proxies for that channel: above-median monthly expenditure, charcoal as the main cooking fuel, LPG as the main cooking fuel, the respondent perceiving electricity as the most expensive fuel, and baseline familiarity with EPCs (columns 1–5). Every interaction is statistically indistinguishable from zero, with point estimates between -1.75pp and $+2.85\text{pp}$. On gender, we interact the information treatment with two measures: the respondent’s gender (whose preferences are elicited) and the household head’s gender (a decision-rights channel). Appendix Table D11 reports both interactions, pooled and by country, with cluster-robust standard errors at the neighborhood level. The pooled treatment $\times$ respondent-female interaction is -5.92pp (SE 3.45, $p \approx 0.086$). The treatment effect is $+4.43\text{pp}$ for male respondents and approximately -1.48pp for female respondents (column 1). However, the pooled treatment $\times$ head-female interaction is essentially zero (0.56pp, SE 2.59; column 5), and by-country estimates of both interactions are null with wide confidence intervals (the largest by-country interaction is -12.72 for respondent-female in Uganda, SE 12.88). One possible explanation is that the script confirmed female respondents’ priors. Women cook and are likelier to know electricity is cheaper than charcoal per meal. Male respondents, less exposed to cooking-budget arithmetic, had more room to update. An alternative is Hawthorne-style updating tied to enumerator interaction rather than message content. We flag two cautions before reading the respondent-female result as substantive. First, the direction is opposite the cooking-budget prior that motivated the test. Second, the marginal significance does not survive a multiple-testing correction. Applying a Romano-Wolf step-down across the family of five heterogeneity interactions (respondent female, head female, charcoal-main, LPG-main, and prior EPC familiarity) returns RW $p = 0.100$ for the respondent-female interaction, exactly at the 10% threshold, and RW $p \geq 0.702$ for each of the other four (0.702 for charcoal-main, 0.761 for prior EPC familiarity, and 0.928 for both head-female and LPG-main). The respondent-female

interaction sits at the boundary of conventional significance once family-wise error rates are controlled, and we do not interpret it as a robust finding.

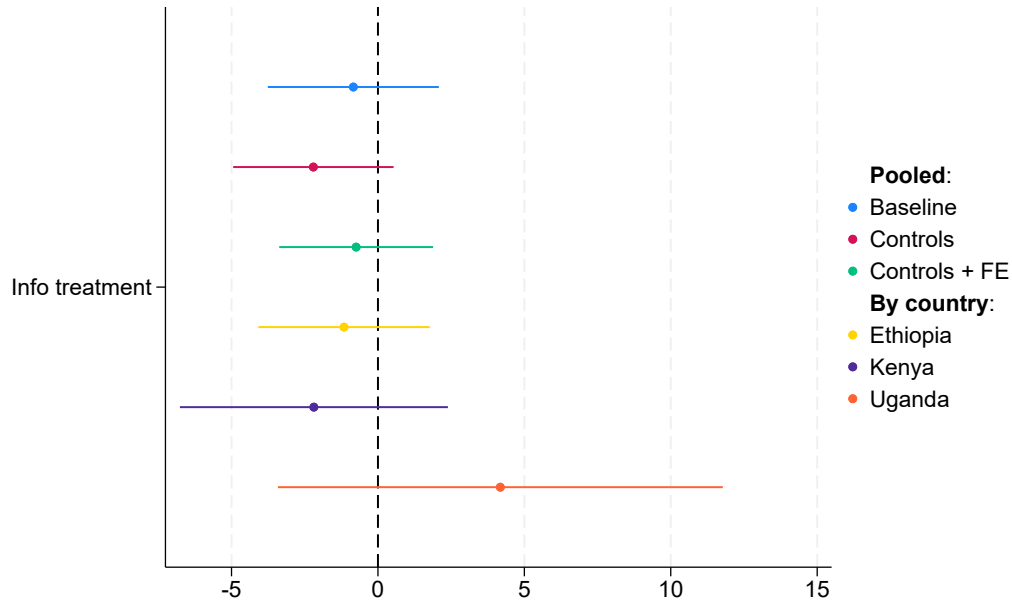
The null is not an artifact of ladder-WTP overestimation. The outcome M_i^{share} is the ladder-elicited WTP, which §5.2 shows overstates revealed demand: 62.1% of at-risk respondents refused the binding offer at a price at or below their stated ladder WTP. We re-estimate the pooled ITT under four adjustments to the ladder outcome (Appendix Table D12): dropping the respondents whose binding refusal exceeded their ladder WTP, dropping those plus the small number of respondents who bought below their ladder WTP, capping ladder WTP at the drawn price (minus 10 pp) for overestimating respondents rather than dropping them, and combining the drop with a high-BDM-understanding restriction. The point estimate moves between -0.35 and -1.37 pp across these adjustments and remains statistically indistinguishable from zero in every column. No adjustment reveals a positive treatment effect that the unadjusted ladder outcome obscures.

One possible explanation for the null is that a non-trivial share of respondents already perceive electricity as a cheap cooking fuel, so the script confirms rather than updates their priors. 25.2% of respondents rank electricity among the two cheapest cooking fuels at baseline (10.9% as cheapest, 14.2% as second-cheapest; Figure 2). However, misperceptions are present in the population. Nearly half of respondents (47.3% pooled; 38.2% in Kenya to 58.2% in Uganda) view electricity as the most expensive cooking fuel, so the design choice to test an information mechanism was not on its face misguided.

A second possible explanation is that our script framed operating-cost savings at the per-meal horizon, which may be too short-horizon to induce updating about a durable purchase. Filippini et al. (2026) run a digital information RCT with motorcycle buyers in Kathmandu, Nepal, in which the same operating-cost savings are presented either at an annual horizon or cumulated over five years. The five-year framing shifts the fuel economy of motorcycles actually purchased by approximately 3.6%, while the annual horizon produces no detectable effect. Message content can matter at smaller stakes: Toledo (2016) finds that a persuasive environmental appeal

raises LED-bulb take-up by 6 percentage points in Brazilian favelas, where the bulb costs a small fraction of the 47–67% of monthly expenditure the EPC commands here. The per-meal framing we used is shorter still than the annual condition in Filippini et al., and the EPC is itself a durable bought once and used for years – a horizon mismatch between the message and the purchase decision the household is being asked to make. We hypothesize that a redesigned information treatment that presents EPC operating-cost savings cumulated over the appliance’s expected lifetime, rather than per meal, would generate a measurably larger WTP response, and we flag this as the natural next step for a follow-up information experiment.

Figure 6: Information-treatment effect on WTP across specifications



Notes: Point estimates and 95% confidence intervals for the information-treatment coefficient on willingness to pay (share of market price, 0–100). Pooled panels (Baseline, Controls, Controls + FE) correspond to columns 1–3 of Table 3. By-country panels (Ethiopia, Kenya, Uganda) correspond to columns 1, 3, and 5 of Appendix Table D3 (robust-SE estimates). Controls: respondent female, household size, main cooking fuel charcoal, main cooking fuel LPG, typical weekly outage hours (winsorized at country-specific p95). “Controls + FE” additionally absorbs country fixed effects. All intervals use heteroskedasticity-robust standard errors; wild-cluster bootstrap p -values for the by-country estimates are reported in Table D3.

6 Concluding Remarks

6.1 Limitations

Four design constraints bound interpretation. First, the survey is cross-sectional: we observe stated and revealed WTP at a single point of sale and cannot track whether BDM buyers use their EPCs, switch back to prior fuels, or influence neighbors to adopt. Long-run adoption and disadoption dynamics, as well as spillovers, remain unmeasured. Second, the sample is restricted to urban, grid-connected households. Rural households and households below MTF Tier 4 service quality are outside the sampling frame, and we cannot extrapolate our subsidy bounds to off-grid or peri-urban populations. Third, the bounds are the honest product of the design, not a shortcut around it. The overstatement classifier is too weak to support household-level correction, the Ethiopia per-protocol gating tilts the binding-only curve toward higher-WTP households, and we cannot rule out strategic responses subtler than bunching at the ladder maximum, so we report a bracket rather than a point estimate of demand. Fourth, the mechanism evidence is diagnostic rather than experimental. The single per-meal information arm cannot distinguish a failure to update beliefs from updated beliefs that do not translate into WTP, and we did not randomize financing, so the liquidity interpretation rests on the payback and refusal patterns rather than on a credit treatment.

6.2 Implications for Policy

EPC adoption in Ethiopia, Kenya, and Uganda will not scale without deep subsidies. At unsubsidized retail prices, predicted take-up is below 5% in all three countries; reaching 50% adoption requires subsidies of 56–90% of retail price. The bracket itself, stated-ladder floor to binding-choice ceiling, is the policy-relevant object. A program that budgets from stated-preference data alone would undershoot the required transfer by up to 29 percentage points in Kenya and Uganda. These are back-of-envelope targets. Our sample covers urban grid-connected households, so national subsidy costs depend on coverage rates and distribution infrastructure. The magnitudes are consistent with the low WTP

documented for LMIC environmental goods by [Greenstone and Jack \(2015\)](#) and with the energy-efficiency gap for clean-energy durables documented by [Berkouwer and Dean \(2022\)](#) and reviewed by [Fowlie and Meeks \(2021\)](#).

The evidence points to financing, not information, as the next constraint to test. Payback within months, refusal that tracks the purchase price rather than the household-specific return, and a price equal to roughly half of monthly expenditure are the signature of a liquidity constraint compounded by non-salient savings (§5.5), and the cheap per-meal information treatment does not move WTP. Credit or rebate-at-purchase designs in the spirit of [Berkouwer and Dean \(2022\)](#), who find that a loan more than doubles WTP for efficient charcoal stoves, are therefore the natural follow-up experiment. However, our design cannot resolve the credit question directly, and information framed over the appliance lifetime rather than per meal remains untested (§5.6).

Our estimates price the announced electric-cooking targets. Kenya’s Pika na Power initiative, Uganda’s National Integrated Clean Cooking Strategy target of 18% household electric cooking by 2040, and Ethiopia’s Electrification Program all depend on household demand far above what we measure. Whether those targets are met by utility cross-subsidy, fiscal transfer, or donor-financed rebates-at-purchase is a fiscal and political question our demand estimates do not resolve. However, the size of the gap, at least 85% of retail in every country on the binding-choice curve, means any credible pathway involves substantial per-unit public expenditure. Monthly expenditure is the strongest within-country demand predictor, which suggests expenditure-targeted subsidies could raise adoption per public dollar relative to uniform price cuts, though our cross-sectional design cannot verify targeting performance.

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A BDM protocol: enumerator script and respondent flow

This appendix documents the BDM protocol verbatim, in the order an enumerator administered it. All scripted text reproduced below is drawn directly from the country-specific CAPI XLSForm instruments. The script is structurally identical across countries. Country-specific differences are limited to the practice good, currency, price ladder, examples of dishes, and information-treatment cost figures. Section 4.1 of the main text reports the empirical implications of the protocol. This appendix supplies the script itself.

A.1 Materials and pre-BDM verification

Each enumerator carried, in addition to the CAPI tablet: (i) one physical EPC unit identical to the model used to set the country-specific reference market price P_c^{market} ; (ii) physical units of the country-specific practice good: petroleum jelly (Uganda), Menengai petroleum bar soap (Kenya), or 1-liter Largo Liquid Soap (Ethiopia); (iii) an opaque black bag with identical tokens (“balls”), one per price tier on the EPC ladder, and a parallel set for the practice-good ladder; and (iv) a cash float and receipt book sized to the country’s expected EPC purchase volume. In Ethiopia and Kenya, enumerators reused the same opaque bag for the practice and EPC draws, swapping the tokens between rounds. In Uganda, the two draws used separate bags. The tokens were table-tennis balls in Ethiopia and folded paper slips in Kenya. The Uganda token medium was not recorded. Ethiopia enumerators showed respondents the price ladder printed in the questionnaire while reading the tiers aloud, and Kenya enumerators carried a printed price-tier card. Whether Uganda enumerators displayed a printed ladder was not recorded.

The BDM module was administered only after the respondent had completed the demographics, cooking-energy, and pre-treatment price-perception modules, and after the practice and EPC goods had been physically placed in front of her. The CAPI logic gated the BDM on a screener variable verifying that the respondent was the household head, spouse of the head, or the primary cook. We gated on these roles to ensure the BDM respondent could plausibly execute a purchase decision in a real market. For the same reason, enumerators allowed the respondent to contact another household decision-maker (for example, to call the household head by phone) before committing to her ladder maximum and before accepting or declining the drawn price. We wanted the elicitation to mirror an actual market transaction as closely as possible. When the eligible respondent was not at home, field teams did not substitute another adult. Ethiopia teams, who visited households with Ethiopian Electric Utility (EEU) staff as guides, collected contact information, made an appointment, and revisited to complete the interview with the eligible respondent. Kenya teams, who sampled along a walking route without a prior listing, moved on to the next eligible household instead. The Uganda field team did not document its rule.

A.2 Framing and incentive explanation

After confirming the practice good and EPC were in view, the enumerator opened the BDM module with the following framing, read verbatim from the CAPI instrument (identical across the three country instruments):

“Dear participant. Thank you for participating in the survey up to this point. I am now going to chat with you about this device. Enumerator: stop now and show the Electric Pressure Cooker (EPC) to the respondent.”

The enumerator next asked whether the respondent recognized the device (“Do you know this device?”; if yes, “What is it?”), checking for correctness and explaining as needed. The enumerator then read the EPC description, with the list of dishes localized to each country:

“Well, this is an Electric Pressure Cooker; it cooks fast, uses little electricity; it can cook almost all dishes that you may want to prepare in your household. Like matooke, posho, beans, meat and frying [Uganda] / ugali, beans, makande, stewed meats, and frying [Kenya] / meat Wot, Kikil, and frying [Ethiopia]. This is because the pressure cooker also conserves the pressure, which it uses to continue cooking while it stops consuming electricity.”

The mechanism was then explained in a single block, identical across the three country instruments except for the placeholder price units. We reproduce the Uganda version:

“Today, we are going to allow you to buy this EPC, at a price that will depend on your responses. I am going to ask questions about the highest price for which you are willing to buy this EPC. I will then ask you to randomly draw one ball from this black bag that contains 10 balls with 10 different prices written on them. If the price written on the ball is below the maximum price you stated, you will be allowed to buy the EPC at the price written on the ball. If the price of the ball is above the maximum price you stated, you not be allowed to buy the EPC from us. The offer to buy the EPC from us is valid for up to one month only. If you are interested in buying the EPC from us, I encourage you to think carefully about the maximum price you are willing to pay for it. Remember, we also carefully thought about the market price of the EPC while deciding on the prices written on the balls in the black bag. I will not show you the prices on the balls, but I confirm to you they are different prices, with one of the balls having the highest price, which is the market price of this EPC (what we bought the EPC at). Do you have any questions about my explanations?”

One feature of this script bears noting. The script states that the bag “contains 10 balls,” but the Ethiopia and Kenya EPC ladders contain 11 tiers (0% through 100% of P_c^{market} in 10pp steps), so the realized bag in those countries contained 11 tokens. The Uganda EPC ladder contains exactly 10 tiers (0 UGX and 30,000–270,000 UGX in 30,000 UGX steps), consistent with the script. The Ethiopia and Kenya field teams confirm that the physical bag in those countries contained 11 tokens, one per ladder tier, and that enumerators verbally corrected the script to “11 balls” when delivering it.

A.3 Practice round: practice-good BDM

To familiarize the respondent with the mechanism before the focal EPC elicitation, the enumerator administered a structurally identical BDM round on a low-stakes practice good (referred to throughout the paper as the jelly BDM, following East African usage). The practice good and ladder were:

- Uganda: petroleum jelly, 10-tier ladder at 0, 1,000, 2,000, ..., 9,000 UGX. Practice question: “Would you buy this petroleum jelly at UGX [tier]?”
- Kenya: Menengai petroleum bar soap, 10-tier ladder at 0, 5, 10, ..., 45 KES. Practice question: “Would you buy this petroleum Menengai bar soap at Ksh. [tier]?”
- Ethiopia: 1-liter Largo Liquid Soap, 10-tier ladder at 0, 16, 32, ..., 144 ETB. Practice question: “Would you buy this 1-liter Largo Liquid Soap (market price up to 50 ETB) at ETB. [tier]?”

The enumerator walked the respondent through each tier in ascending order. After the final tier, the CAPI instrument computed the highest tier the respondent had accepted and displayed it to the enumerator as “Maximum price to buy the [practice good].” The enumerator then instructed the respondent to draw the binding price:

“Please now draw one ball to decide on the price for the petroleum jelly.”

The enumerator recorded the drawn price (restricted by the CAPI choice list to the country-specific practice-good ladder), and the CAPI instrument auto-determined the transaction outcome via two pre-written notes the enumerator read to the respondent:

“At this drawn price, the respondent has the opportunity to buy the [practice good].”

“At this drawn price, the respondent does not have the opportunity to buy the [practice good].”

If the drawn price was at or below the respondent’s stated maximum, the respondent paid in cash and received the practice good on the spot. Otherwise no transaction occurred.

A.4 EPC BDM elicitation

The EPC BDM followed the practice round. The price ladder was country-specific:

- Ethiopia: 11 tiers at 0, 1,150, 2,300, ..., 11,500 ETB (i.e., 0, 10, ..., 100% of $P_{ET}^{\text{market}} = 11,500$ ETB).
- Kenya: 11 tiers at 0, 1,075, 2,150, ..., 10,750 KES (i.e., 0, 10, ..., 100% of $P_{KE}^{\text{market}} = 10,750$ KES).
- Uganda: 10 tiers at 0 UGX and 30,000, 60,000, ..., 270,000 UGX (i.e., 0% and 11.1%, 22.2%, ..., 100% of $P_{UG}^{\text{market}} = 270,000$ UGX).

The enumerator walked the respondent through each tier: “Would you buy this EPC at [tier]?”. The CAPI instrument computed the maximum tier and displayed it to the enumerator as “Maximum price to buy the EPC,” and the enumerator administered the binding draw:

“Please now draw one ball to decide on the price for the EPC.”

The enumerator recorded the drawn price (restricted by the CAPI choice list to the EPC ladder), and the CAPI instrument determined the transaction outcome via the same pre-written rule used in the practice round: “At this drawn price, the respondent has [does not have] the opportunity to buy the EPC.” If the drawn price was at or below the respondent’s stated maximum, the respondent paid in cash and received the EPC. Otherwise, no transaction occurred.

Field-team reports confirm that the on-the-spot transaction was the norm in all three countries. Most buyers paid cash and received the physical EPC at the household visit. Uganda teams traveled with one EPC and a driver who promptly replaced each unit sold. In Ethiopia, three late purchases could not be fulfilled on the spot and were delivered to the household by a project driver within days. Delayed payment under the one-month window was the exception, used by the few buyers facing a cash shortage. Delayed buyers contacted the original enumerator in Kenya and Uganda and the research team in Ethiopia. The EPC was then delivered to the household in Ethiopia and Kenya, while Uganda buyers collected it from the research team at Makerere University.

A.5 Informal gating of zero-ladder respondents (Ethiopia)

In Ethiopia, enumerators did not administer the binding draw to respondents who set their ladder maximum $M_i = 0$. The CAPI form does not contain skip logic enforcing this, but field-team records indicate that the gating was systematic: 114 of 501 Ethiopia respondents

(22.75%) have no recorded binding draw, and all 114 have $M_i = 0$. 101 of these were initially miscoded in the raw data as $P_i = \text{"Etb 0"}$ (i.e., the bottom tier of the ladder) and were recoded to missing in `1_datacleaning.do`. The remaining 13 were missing in the raw data. The Ethiopia field team confirms, after checking with supervisors and enumerators, that the gating was a uniform practice adopted by common agreement across all enumerators, motivated by a wish not to embarrass respondents who stated a maximum of zero. The team reports no other informal screening rules at the BDM stage. The corresponding non-participation rates in Kenya ($1/537 = 0.19\%$) and Uganda ($4/268 = 1.49\%$) are negligible. Per-protocol estimates of $\Pr(\text{buy}_i = 1 \mid P_i^{\text{share}})$ exclude the 119 non-participants and therefore overstate Ethiopia demand at the bottom of the price ladder. Intent-to-treat regressions on M_i^{share} use the full sample ($N = 1,306$). Per-protocol regressions on buy_i use the BDM-participant subsample ($N = 1,187$).

A.6 Information-treatment integration

For respondents randomized into the information arm, the CAPI instrument routed through a parallel set of BDM nodes, structurally identical to the control-arm flow except for an additional information module inserted between the EPC description and the BDM mechanism explanation. The information module begins with the heading “Extra Information!!!” and proceeds as a short structured Q&A. For Uganda the script reads:

“Extra Information!!! If you are cooking 1 Kg of dry beans, how much worth of charcoal do you use?” [enumerator records the respondent’s amount] “For how long?” [enumerator records minutes]
 “For the EPC however, you will consume about UGX 500 worth of units only and cooking of about 1 hour. This is because the pressure cooker also conserves the pressure, which it uses to continue cooking while it stops consuming electricity.”

The Kenya script is identical except for the currency and price change. The Ethiopia script substitutes a country-relevant dish:

“If you are cooking meat sauce (meat, wot, kikil), how much worth of charcoal do you use? For how long?”

Delivery mode differed somewhat across countries. In Ethiopia, enumerators delivered the operating-cost comparison verbally, supported by summary materials prepared for the enumerators. Respondents were not shown the underlying source document. In Kenya, enumerators delivered the comparison both verbally and with a printed one-page summary shown to the respondent, rather than the full eCookbook. Both teams confirm that the

content read in the field matched the script reproduced above. The Uganda team did not document its delivery mode beyond the CAPI script itself.

After the information module, the CAPI instrument proceeded through the same mechanism explanation, practice-good BDM, and EPC BDM as the control arm. Inclusion of the practice-good BDM after the information module, rather than before, is a CAPI-form decision. The practice good itself is not the object of the treatment, but its placement after the information module means treated respondents practiced the mechanism with the information already delivered.

A.7 Post-BDM debrief

Immediately after the EPC binding outcome the CAPI instrument recorded four enumerator-administered debrief items (identical across the three country instruments):

- “Did the respondent want to change their mind?”
- “Did the respondent argue?”
- “What was the respondent’s argument?” [open-text]
- “Enumerator: Did the respondent buy the EPC?”

We use the two yes/no debrief items (“Did the respondent want to change their mind?” and “Did the respondent argue?”) as qualitative inputs to the consistency-corrected robustness analyses (Tables D15–D17). The enumerator’s recorded purchase outcome is cross-checked against the CAPI-derived binding outcome (drawn price $\leq$ stated maximum) for internal consistency. We use the CAPI-derived value as the primary buy_i outcome.

A.8 Translation

The CAPI instrument was authored in English, and the tablet displayed English text throughout. Translated scripts were not embedded in the CAPI form. Enumerators administered the survey in Amharic (Ethiopia), in Swahili or English (Kenya), and mainly in Luganda, with a few interviews in English (Uganda). No country produced a formal written translation or back-translation. In Ethiopia, experienced bilingual enumerators and supervisors (minimum BA degree, fluent in English and Amharic) rendered the English questions into natural Amharic in real time. Consistency was established through intensive training, mock-questionnaire sessions, and researcher checks of comprehension during and after training. In Kenya, enumerators translated from English to Kiswahili and back-translated responses on the fly, after training and mock interviews in both languages. In Uganda, bilingual enumerators asked the questions directly in Luganda, and the enumerator training itself walked through the questions in Luganda.

B BDM price exogeneity

The realized BDM offer is randomly assigned by protocol. Each respondent drew her binding price physically from an opaque bag containing identical tokens, one per price tier on the country-specific ladder, after reporting the highest tier she was willing to pay (§4.1). We verify orthogonality empirically because the realized marginal distribution of draws is non-uniform within country (Figure B1), which could in principle reflect enumerator or respondent deviations from the protocol correlated with household covariates rather than sampling variability alone.

We test exogeneity by regressing the realized price share on the household covariate vector:

$$P_i^{\text{share}} = \alpha + V_i' \beta + \gamma_{c(i)} + u_i, \quad (9)$$

where P_i^{share} is the drawn BDM offer divided by the country-specific market price, V_i is the vector of household covariates listed in Table B1, and $\gamma_{c(i)}$ are country fixed effects. We test $H_0 : \beta = 0$ jointly using both the cluster-robust F -statistic and a wild-cluster bootstrap (Webb weights, 9,999 replications). We treat the bootstrap as the preferred inference (§4.6). Failure to reject H_0 confirms the price draw was randomized as intended.

Table B1 reports the per-covariate coefficients and joint tests pooled and by country; Figure B1 shows the empirical distribution of price draws within country.

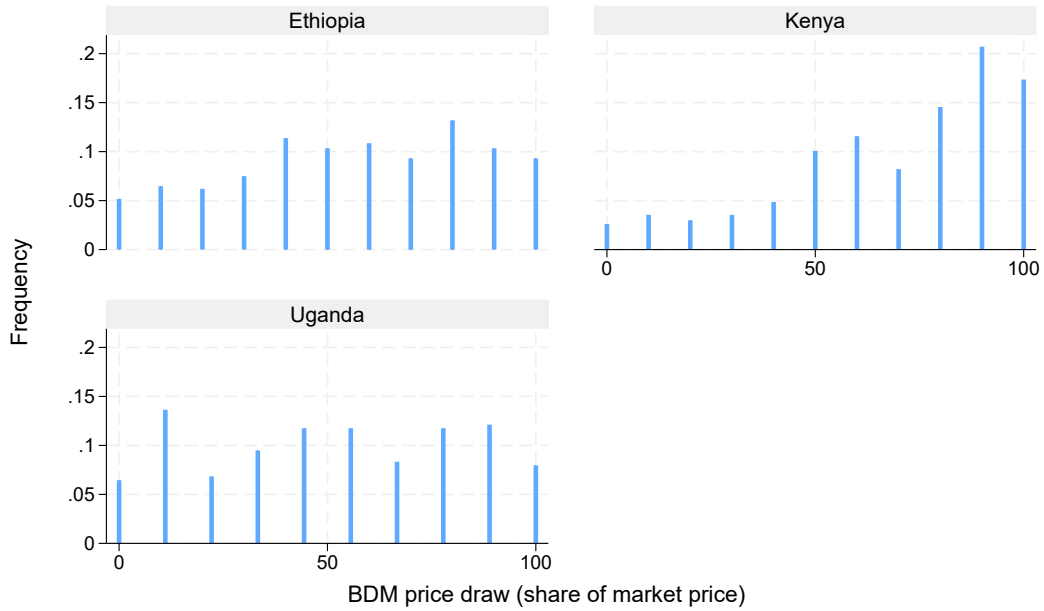
Table B1: Realized BDM price is orthogonal to household covariates, conditional on country

	(1) Pooled + Country FE	(2) Ethiopia	(3) Kenya	(4) Uganda
Respondent female	3.2139 (2.5202)	7.1353 (4.1204)	-6.2440 (3.7011)	8.5912 (6.5692)
Respondent age	-0.0809 (0.0890)	-0.1497 (0.1605)	-0.0410 (0.2050)	-0.0393 (0.1333)
Respondent is HH head or spouse	0.9988 (3.3155)	4.4653 (4.8252)	-9.2205 (6.0712)	3.7055 (7.4503)
Impatience score (# today, 0–6)	0.1260 (0.3946)	0.5734 (0.8909)	-0.3438 (0.4136)	0.4687 (1.2667)
Variance tolerance (# variable, 0–6)	0.8047 (0.5553)	0.4636 (1.6301)	0.7508 (0.6142)	1.2816 (1.2874)
HH head female	-4.8616 (3.4183)	-5.3645 (6.5244)	-0.3755 (3.4747)	-6.4037 (7.2602)
Household size	-0.0932 (0.5535)	0.4561 (1.2181)	-1.0192* (0.5801)	0.0551 (0.9102)
Primary cook female	-5.3589 (3.9515)	1.6691 (14.8347)	1.3042 (5.6226)	-14.4699** (6.9510)
Primary cook age	0.0020 (0.0963)	0.0217 (0.1572)	-0.0469 (0.1895)	-0.1624 (0.1925)
Primary cook is spouse of HH head	-2.7951 (2.9478)	-3.9929 (5.5586)	1.2172 (3.0609)	-1.9170 (8.5081)
Monthly expenditure (asinh, USD)	-1.0021 (1.1047)	-0.6692 (2.5760)	-0.9155 (1.3318)	-4.0761 (2.7195)
Familiar with EPC	-1.5301 (1.9116)	-0.8508 (3.2650)	-2.3586 (3.0589)	1.0290 (4.2903)
Cooks with charcoal	2.2745 (2.2872)	8.1513* (4.1547)	2.6638 (2.3895)	-7.3474 (10.3439)
Cooks with LPG	3.8073 (3.2851)	10.4406* (5.1379)	7.5241 (6.7119)	-3.9856 (3.9284)
Cooks with electricity	-2.0657 (3.4096)	2.7975 (10.8060)	-1.2087 (4.9751)	-1.6716 (4.8165)
Cooks with firewood	0.1541 (2.1268)	-6.0377** (2.5626)	2.0468 (3.6422)	4.6696 (4.4555)
Grid is main elec. source	-2.2594 (2.5758)	0.0000 (.)	-2.0085 (2.6258)	-4.3620 (6.6294)
Typical outage hrs/week	0.0693 (0.0740)	0.0069 (0.0782)	-0.0187 (0.1726)	0.4130 (0.2469)
Joint F (covariates)	1.44	3.01	13.77	4.19
Joint F p-value (cluster-robust)	0.147	0.034	0.000	0.001
Joint F p-value (wild-cluster bootstrap)	0.601	0.999	0.280	0.575
Observations	1186	386	536	264
R ²	0.080	0.040	0.024	0.064

Notes: OLS estimates of the realized BDM price draw (in % of market price, 0–100) on household covariates from Table 1. Column 1 pools the three countries with country fixed effects (suppressed); columns 2–4 report by-country estimates. Sample restricted to respondents who drew a BDM price; in Ethiopia, 114 respondents who stated a willingness to pay of zero in the ladder were not offered a draw under the enumerator protocol (Appendix A). Standard errors clustered at the neighborhood level (63 clusters total: 13 in Ethiopia, 26 in Kenya, 24 in Uganda); sample sizes 387, 536, and 264 households respectively. The wild-cluster bootstrap p -value (Webb weights, 9,999 replications, seed 20260508) is the preferred small-sample inference for the joint null that all covariate coefficients equal zero; cluster-robust F -statistics are reported alongside but are unreliable with fewer than thirty clusters per country. The variable `elec_outages_dur_w` is typical outage hours per week, winsorized at country-specific 95th percentiles (caps of 63 hours/week in Ethiopia, 30 hours/week in Kenya and Uganda); households reporting zero outages were not asked the duration question and are coded as zero hours. Significance:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B1: BDM price draws are non-uniform within country



Notes: Histogram of realized BDM price draws as share of country-specific market price, by country. Ethiopia ($N = 387$) and Kenya ($N = 536$) draw from an 11-tier grid (0%, 10%, ..., 100%); Uganda ($N = 264$) draws from a 10-tier grid (0%, 11.1%, 22.2%, ..., 100%). The Ethiopia sample excludes 114 respondents who were not offered a draw under the enumerator protocol because they stated a willingness to pay of zero in the ladder (Appendix A). Chi-square goodness-of-fit tests against discrete uniformity: Ethiopia $\chi^2(10) = 26.1$, Kenya $\chi^2(10) = 235.6$, Uganda $\chi^2(9) = 112.8$; all reject at the 1% level. The non-uniformity is driven by a high-price skew in Kenya (over-draws at 80–100% of market price) and milder lumpiness in Ethiopia and Uganda. All three countries under-draw the zero tier. Identification of the demand schedule does not depend on uniformity of the draw, only on its orthogonality to household covariates conditional on country (Table B1).

C Correlates of stated willingness to pay

This appendix characterizes who states high WTP in the first place. Expenditure, baseline fuel use, and grid access explain most WTP variation. To assess this we regress maximum stated WTP, expressed as a share of market price (0–100), on demographic, expenditure, fuel-use, decision-rights, grid-access, and reliability covariates, pooling 1,304 households across the three countries.¹⁹ Table C1 reports OLS estimates with robust standard errors. Columns 1–3 give country-specific estimates, column 4 pools the three samples, and column 5 adds country fixed effects. Across the correlates family of four pooled+FE coefficients we interpret in the text – expenditure, head female, prior EPC familiarity, and outage hours (§4.6) – two members survive a Romano-Wolf step-down correction (Romano & Wolf, 2005; Clarke et al., 2020) for the family-wise error rate: expenditure at the 1% level (RW $p < 0.001$) and prior EPC familiarity at the 10% level (RW $p = 0.088$). Head female and outage hours do not survive (RW $p = 0.558$ and 0.985 , respectively).

Five theoretical channels motivate the covariates we include, and we organize the discussion by channel.

Household budget and short-run liquidity constrain lump-sum durable purchases when credit is incomplete (Berkouwer & Dean, 2022; Gill-Wiehl et al., 2025). Monthly expenditure is the strongest single correlate of WTP: doubling monthly expenditure is associated with a 2.86 percentage-point increase in WTP (s.e. 0.63) in the pooled specification with country fixed effects, significant at the 1% level.²⁰ The association holds country by country in Ethiopia (2.22 pp, s.e. 0.82) and Kenya (3.16 pp, s.e. 0.92), and is positive but imprecisely estimated in Uganda (2.68 pp, s.e. 1.64). One possible explanation is that higher-expenditure households also have more cash on hand on the day of the BDM exercise, easing a short-run liquidity constraint that would otherwise bind on a lump-sum purchase. The pattern is consistent with the credit-constraint mechanism documented for Nairobi charcoal stoves, where households facing a 295% private return are nonetheless willing to pay only \$12 absent a loan offer (Berkouwer & Dean, 2022). However, expenditure also proxies for permanent income, household composition, and unobserved taste for modern appliances, so we cannot isolate the liquidity channel from these alternatives with cross-sectional data.

¹⁹We omit respondent gender and a respondent-is-head-or-spouse indicator because both have limited within-sample variation – the respondent is almost always the head or spouse, so these variables track who happened to be home rather than household decision rights. Household-head gender is included separately and captures the decision-rights channel.

²⁰The table reports the coefficient on inverse-hyperbolic-sine monthly expenditure (4.13 pp per unit), which behaves like a log transformation at the values we observe. We multiply by $\ln(2) \approx 0.693$ to express the magnitude as a per-doubling association. The country-specific magnitudes in the next sentence are converted the same way.

Current fuel use proxies the opportunity cost of switching: households on recurrent-cost fuels stand to gain most from a substitute, and across LMICs income, fuel availability, and prices are the determinants of clean-stove adoption most consistently identified in systematic reviews (Lewis & Pattanayak, 2012; Jeuland et al., 2021; Guta et al., 2022). The primary cooking fuel splits households into LPG buyers, who are willing to pay more, and charcoal buyers, who are willing to pay less. We code each household’s primary stove from the single fuel it names as its main cooking technology, then enter indicators for “main fuel is charcoal” and “main fuel is LPG” alongside the other covariates. The omitted reference category is any other primary fuel (mostly electricity in Ethiopia, firewood elsewhere). In the pooled+FE specification, LPG-main households have WTP 4.90 pp higher (s.e. 2.19, $p < 0.05$) than the within-country reference category, while charcoal-main households have WTP 4.40 pp lower (s.e. 1.68, $p < 0.01$). The country-specific estimates anchor the pooled+FE pattern: in Ethiopia, charcoal-main households (the 17% who stack charcoal on top of an electric mitad) have WTP 6.60 pp lower (s.e. 1.48, $p < 0.01$) than otherwise-comparable electric-main households. In Kenya, LPG-main households have WTP 8.05 pp higher (s.e. 3.24, $p < 0.05$) than the small firewood/other-main residual.²¹ One possible explanation for the LPG-main premium is that LPG-using households have already crossed the upfront fixed cost of acquiring a clean recurrent-cost cooker, demonstrating both willingness and means to pay for clean cooking. For charcoal-main households, the upfront cost is the binding margin and the EPC arrives as a second large lump-sum demand.

Electricity access and reliability bound the technology’s usable value (Lee, Miguel, & Wolfram, 2020; Meeks & Pokhrel, 2024). Grid access is associated with WTP. Outage exposure is not. Grid access is associated with a 5.29 pp higher WTP (s.e. 2.69, $p < 0.05$) in pooled+FE, although only 76 households (5.82%) are off-grid so this estimate is identified off a small subsample. The pooled outage coefficient of -0.23 pp per hour (s.e. 0.046, $p < 0.01$) collapses to -0.01 (s.e. 0.046) once country fixed effects are added, and by-country outage estimates are indistinguishable from zero.²² The pooled correlation reflects cross-country variation – Ethiopia reports both the highest outage exposure (19.8 hours/week) and lower mean WTP

²¹Pooled-without-FE, the charcoal coefficient is +6.45 pp (s.e. 1.79, $p < 0.01$) because charcoal-main households are concentrated in Uganda, the highest-WTP country (§5.1). Country fixed effects absorb this between-country compositional channel and isolate the within-country association, which flips negative. The pooled-without-FE estimate therefore conflates two distinct margins. The FE estimate is the relevant one for predicting WTP within a given urban market.

²²By-country outage estimates: Ethiopia -0.006 (s.e. 0.046); Kenya -0.081 (s.e. 0.146); Uganda 0.195 (s.e. 0.166). The within-country null is unchanged when outage exposure is measured as the count of outages in a typical week or as a binary indicator for any outages, rather than as winsorized hours (results available on request). Pairwise correlations among the three measures range from 0.39 to 0.69 (all $p < 0.001$).

Table C1: Correlates of willingness to pay for an EPC

	(1) Ethiopia	(2) Kenya	(3) Uganda	(4) Pooled	(5) Pooled + FE
Respondent age	-0.084 (0.066)	-0.007 (0.184)	0.009 (0.196)	-0.071 (0.071)	-0.075 (0.066)
HH head female	0.749 (1.565)	-2.676 (2.473)	-0.738 (3.751)	-0.312 (1.432)	-1.021 (1.365)
Household size	0.221 (0.377)	0.460 (0.681)	0.080 (0.642)	-0.068 (0.313)	0.210 (0.308)
Primary cook age	-0.042 (0.078)	-0.036 (0.188)	-0.228 (0.206)	-0.024 (0.077)	-0.024 (0.073)
Primary cook is hired help	-3.069 (2.656)	29.071*** (7.813)	-7.482 (10.009)	-4.751 (3.159)	0.351 (2.913)
Monthly expenditure (asinh, USD)	3.310*** (1.141)	4.277*** (1.346)	4.583* (2.475)	5.546*** (0.978)	4.026*** (0.895)
Familiar with EPC	2.658 (2.172)	4.712** (2.347)	-2.015 (4.175)	4.012** (1.600)	2.330 (1.549)
Main cooking fuel: charcoal	-6.603*** (1.477)	-0.469 (5.601)	-5.879 (4.455)	6.450*** (1.788)	-4.399*** (1.684)
Main cooking fuel: LPG	1.148 (3.800)	8.045** (3.235)	-4.401 (6.784)	15.344*** (1.776)	4.897** (2.187)
Grid is main elec. source	0.000 (.)	3.803 (3.527)	6.872 (4.778)	1.664 (2.775)	4.640* (2.759)
Typical outage hrs/week	-0.003 (0.045)	-0.001 (0.147)	0.187 (0.178)	-0.211*** (0.046)	-0.001 (0.045)
Constant	4.054 (6.966)	6.268 (10.084)	20.581 (16.794)	-1.824 (6.705)	-6.037 (6.240)
Observations	499	537	268	1304	1304
R ²	0.064	0.084	0.042	0.169	0.248

Notes: OLS estimates. Outcome is WTP as share of market price (0–100). Robust standard errors in parentheses. Columns 1–3 are country-specific samples; columns 4–5 pool the three countries, with column 5 adding country fixed effects (coefficients suppressed). Typical outage hours per week is winsorized at country-specific 95th percentiles (caps of 63 hours/week in Ethiopia, 30 hours/week in Kenya and Uganda); households reporting zero outages were not asked the duration question and are coded as zero hours. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

than Kenya. One possible explanation for the within-country null is that households substitute toward backup arrangements (charcoal, LPG retention) that smooth the reliability cost.

Information and prior exposure reduce learning costs (Mahadevan et al., 2025; Pattanayak et al., 2018). Prior EPC familiarity enters with a positive but imprecise coefficient: 2.33 pp pooled+FE (s.e. 1.55), significant in Kenya alone (4.71 pp, s.e. 2.35, $p < 0.05$).²³ One possible explanation is that prior exposure lowers cognitive switching costs (Jeuland et al., 2021), but the imprecision precludes a stronger conclusion.

Intrahousehold decision rights shape adoption when the cook and household head differ in preferences (Jack, Jayachandran, Malagutti, & Rao, 2024). However, household demographics are not associated with WTP in our data. The pooled+FE coefficient on household-head gender is -1.38 pp (s.e. 1.35), and on the primary cook being hired help is 0.18 pp (s.e. 2.91). Coefficients on respondent age, primary-cook age, and household size are also small and imprecise (Table C1).²⁴ One possible explanation is design: our BDM elicits a household-level WTP rather than person-specific WTP, so we cannot detect cook-versus-head divergence even if one exists (Jack et al., 2024).

The correlates regression uses the continuous ladder WTP as its outcome, which overstates revealed demand: §5.2 documents that 62.1% of at-risk respondents refused the binding offer at a price at or below their stated ladder WTP. We therefore re-estimate equation (7) under four adjustments to the ladder outcome – dropping binding-refusal overestimators, dropping those plus the two binding-purchase underestimators, capping overestimators’ ladder WTP at the drawn price (minus 10 pp) rather than dropping them, and combining the drop with a high-BDM-understanding restriction (Appendix Table D9). The three largest correlates retain their direction and significance across all four adjustments. First, monthly expenditure stays positive at the 1% level (3.87–3.97 pp per asinh unit vs. 4.03 in the unadjusted column). Second, main-fuel charcoal stays negative at the 1% level (-4.44 to -5.52 pp vs. -4.40 in the unadjusted column). Third, main-fuel LPG stays positive ($+3.74$ to $+5.44$ pp, $p < 0.05$ in the cap-WTP column and $p < 0.10$ in the three drop-based columns, vs. $+4.90$ in the unadjusted column at $p < 0.05$). The outage and demographic nulls also persist. The one coefficient that attenuates sharply is grid access, which falls from 4.64 pp ($p < 0.10$) in the unadjusted column to 2.37–3.36

²³The 4-correlate Romano-Wolf step-down (§4.6) returns RW $p = 0.088$ for prior EPC familiarity, which clears the 10% threshold but not the 5% threshold. The unadjusted analytical p -value from the heteroskedasticity-robust SE in Table C1 is approximately 0.13. We treat the result as suggestive rather than robust.

²⁴Elicited time and variance preferences also do not predict WTP. Adding the impatience and variance-tolerance counts from Module I (Table 1) to the pooled+FE specification leaves every other coefficient unchanged and moves R^2 by less than 0.001 (0.248 in both columns). The two preference coefficients are 0.22 pp (s.e. 0.30) and -0.29 pp (s.e. 0.40) per unit on the 0–6 scale, both indistinguishable from zero. We therefore omit them from Table C1.

pp (all ns) across the four adjustments. The pattern is consistent with off-grid households being disproportionately represented among the ladder-overestimators identified in §5.2, so the unadjusted grid coefficient partly reflects inflated stated WTP rather than revealed-preference variation. The fuel and expenditure associations are robust to this adjustment.

D Additional Tables & Figures

Figure D1: Representative electric pressure cooker models sold in Ethiopia, Kenya, and Uganda



Table D1: Balance: information-treatment vs. control on baseline covariates

	All			Ethiopia			Kenya			Uganda		
	Treat	Control	Diff	Treat	Control	Diff	Treat	Control	Diff	Treat	Control	Diff
Respondent female	0.85	0.89	-0.037*	0.81	0.85	-0.047	0.89	0.94	-0.051	0.85	0.86	-0.019
Respondent age	44.68	43.74	0.934	46.00	45.87	0.138	43.35	42.57	0.775	45.38	42.34	3.044
Respondent is HH head or spouse	0.90	0.90	0.007	0.85	0.83	0.017	0.95	0.95	0.005	0.88	0.91	-0.032
Impatience score (# today, 0–6)	4.16	4.29	-0.132	3.97	3.80	0.165	4.22	4.68	-0.455**	4.48	4.45	0.031
Variance tolerance (# variable, 0–6)	1.67	1.59	0.077	1.31	1.19	0.119	1.84	1.58	0.262	2.10	2.17	-0.079
HH head female	0.37	0.37	-0.000	0.38	0.40	-0.020	0.33	0.30	0.029	0.48	0.42	0.052
Household size	4.66	4.89	-0.230*	5.06	5.21	-0.147	4.16	4.38	-0.215	5.19	5.14	0.049
Primary cook female	0.96	0.97	-0.010	0.98	0.99	-0.005	0.95	0.98	-0.026	0.92	0.93	-0.013
Primary cook age	41.01	39.58	1.433	40.93	38.82	2.116	41.86	41.57	0.288	38.37	37.91	0.461
Primary cook is spouse of HH head	0.50	0.49	0.008	0.41	0.38	0.033	0.61	0.65	-0.038	0.37	0.43	-0.060
Monthly expenditure (asinh, USD)	5.73	5.67	0.063	5.52	5.51	0.017	5.87	5.79	0.080	5.85	5.73	0.119
Familiar with EPC	0.34	0.33	0.012	0.19	0.19	-0.006	0.51	0.53	-0.023	0.23	0.25	-0.024
Cooks with charcoal	0.77	0.84	-0.066**	0.93	0.94	-0.006	0.58	0.67	-0.093	0.98	0.93	0.047*
Cooks with LPG	0.52	0.42	0.100***	0.13	0.09	0.038	0.93	0.94	-0.011	0.25	0.17	0.082
Cooks with electricity	0.43	0.46	-0.023	0.93	0.97	-0.040	0.05	0.03	0.022	0.30	0.31	-0.012
Cooks with firewood	0.25	0.26	-0.006	0.23	0.22	0.005	0.24	0.28	-0.036	0.35	0.28	0.068
Grid is main elec. source	0.94	0.95	-0.011	1.00	1.00	0. (collin)	0.89	0.91	-0.019	0.89	0.92	-0.026
Typical outage hrs/week	10.28	11.74	-1.456*	18.59	20.87	-2.279	5.19	5.98	-0.791	3.83	6.80	-2.969*
Joint F (covariates)	1.80			114.20			15.95			0.93		
Joint F p-value (cluster-robust)	0.046			0.000			0.000			0.558		
Joint F p-value (wild-cluster bootstrap)	0.475			0.341			0.233			0.998		
Observations	1304			499			537			268		

Notes: Each cell triplet reports treatment-group mean, control-group mean, and the difference. Differences come from regressions of each covariate on the info-treatment indicator with standard errors clustered at the neighborhood level (63 clusters total: 13 in Ethiopia, 26 in Kenya, 24 in Uganda). Per-row stars use the wild-cluster bootstrap p -value (Webb weights, 9,999 replications, seed 20260508); cluster-robust inference alone over-rejects with so few clusters per country. The footer rows report the joint F statistic from regressing the info-treatment indicator on the full covariate vector, with both cluster-robust and wild-cluster bootstrap p -values for the joint null that all covariate coefficients equal zero. Ethiopia's `elec_source_grid` drops in the by-country regressions because the variable is collinear (all sampled Ethiopian households are grid-connected). Typical outage hours per week is winsorized at country-specific 95th percentiles (caps of 63 hours/week in Ethiopia, 30 hours/week in Kenya and Uganda); households reporting zero outages were not asked the duration question and are coded as zero hours. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D2: Cluster (neighborhood) diagnostics

Country	# clusters	Mean HH/cluster	Median HH/cluster	Range
Ethiopia	13	38.5	33	[1, 69]
Kenya	26	20.7	19.5	[1, 80]
Uganda	24	11.2	12	[6, 13]
All	63	20.7	12	[1, 80]

Notes: Cluster identifiers: `district_eth` (Ethiopia, admin level 3), `location` (Kenya, ward), `parish_ward` (Uganda, admin level 3). Range reports minimum and maximum cluster sizes. Ethiopia's 13 clusters fall well below standard cluster-asymptotics rules of thumb; we report wild-cluster bootstrap p -values for country-level inference (Cameron et al., 2008; MacKinnon & Webb, 2017) (Table D3).

Table D3: Effect of information treatment on WTP – by country, with wild-cluster bootstrap inference

	(1) ET (rob)	(2) ET (cl+WCB)	(3) KE (rob)	(4) KE (cl+WCB)	(5) UG (rob)	(6) UG (cl+WCB)
Info treatment	-1.158 (1.490)	-1.158 (0.952)	-2.188 (2.329)	-2.188 (2.285)	4.180 (3.858)	4.180 (4.426)
Respondent female	-0.725 (2.269)	-0.725 (2.077)	-2.321 (3.993)	-2.321 (4.778)	0.262 (4.760)	0.262 (4.752)
Household size	0.545 (0.335)	0.545 (0.388)	0.833 (0.693)	0.833 (0.899)	0.609 (0.625)	0.609 (0.490)
Main cooking fuel: charcoal	-7.455*** (1.485)	-7.455*** (1.299)	-2.264 (5.641)	-2.264 (7.176)	-3.779 (4.474)	-3.779 (4.959)
Main cooking fuel: LPG	1.277 (3.746)	1.277 (3.588)	9.519*** (3.219)	9.519*** (2.918)	1.304 (6.639)	1.304 (7.415)
Typical outage hrs/week	-0.008 (0.045)	-0.008 (0.044)	-0.048 (0.145)	-0.048 (0.125)	0.140 (0.180)	0.140 (0.266)
Constant	16.815*** (2.946)	16.815*** (3.037)	35.853*** (5.170)	35.853*** (6.133)	37.628*** (7.083)	37.628*** (5.789)
Observations	499	499	537	537	268	268
R^2	0.034	0.034	0.028	0.028	0.014	0.014
Wild-cluster p (info treat)		0.330		0.376		0.373

Notes: OLS estimates of the information treatment on WTP (share of market price, 0–100), country by country. “rob” columns report heteroskedasticity-robust standard errors. “cl+WCB” columns report cluster-robust standard errors at the neighborhood level, with the wild-cluster bootstrap p -value (Roodman et al., 2019) (Rademacher weights, 9,999 replications) reported below R^2 . Cluster counts: Ethiopia 13, Kenya 26, Uganda 24. Controls (suppressed): respondent female, household size, main cooking fuel is charcoal, main cooking fuel is LPG (reference category: any other primary fuel), typical weekly outage hours (winsorized at country-specific p95). Significance stars on point estimates correspond to the standard error in the same column. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D4: Ladder demand schedule: subsidy required to reach a given adoption rate, by country (hypothetical, full sample)

Panel A: Ethiopia			Panel B: Kenya			Panel C: Uganda		
Subsidy (%)	Price (share of market)	Share willing to buy	Subsidy (%)	Price (share of market)	Share willing to buy	Subsidy (%)	Price (share of market)	Share willing to buy
100.0	0.000	0.97	100.0	0.000	1.00	100.0	0.000	0.99
90.0	0.100	0.75	90.0	0.100	0.97	88.9	0.111	0.95
80.0	0.200	0.46	80.0	0.200	0.86	77.8	0.222	0.79
70.0	0.300	0.24	70.0	0.300	0.70	66.7	0.333	0.59
60.0	0.400	0.12	60.0	0.400	0.55	55.6	0.444	0.43
50.0	0.500	0.06	50.0	0.500	0.42	44.4	0.556	0.32
40.0	0.600	0.04	40.0	0.600	0.30	33.3	0.667	0.21
30.0	0.700	0.02	30.0	0.700	0.22	22.2	0.778	0.15
20.0	0.800	0.02	20.0	0.800	0.14	11.1	0.889	0.11
10.0	0.900	0.01	10.0	0.900	0.11	0.0	1.000	0.08
0.0	1.000	0.00	0.0	1.000	0.06			

Notes: Each row reports the share of households at each price point in the country-specific BDM ladder who indicate they would purchase an EPC. The “Subsidy” column is the implied per-unit subsidy as a percentage of market price; the “Price” column is the EPC price as a share of market. Ethiopia and Kenya use an 11-tier ladder in 10 pp steps (0, 10, ..., 100% of market). Uganda uses a 10-tier ladder in ~11.1 pp steps (0, 30k, 60k, ..., 270k UGX, where 270k UGX = 100% of market). These shares come from the hypothetical ladder elicitation; the binding BDM-revealed schedule is in Table D5 and, in the main text, Figure 3.

Table D5: Demand schedule from binding BDM choices, by country
Panel A: Ethiopia Panel B: Kenya Panel C: Uganda

Subsidy (%)	Empirical share (ITT)	Probit share (ITT)	Subsidy (%)	Empirical share (ITT)	Probit share (ITT)	Subsidy (%)	Empirical share (ITT)	Probit share (ITT)
100.0	0.77	0.69	100.0	0.93	0.82	100.0	0.93	0.66
90.0	0.37	0.49	90.0	0.74	0.66	88.9	0.47	0.50
80.0	0.19	0.22	80.0	0.25	0.47	77.8	0.44	0.33
70.0	0.11	0.05	70.0	0.21	0.28	66.7	0.00	0.20
60.0	0.00	0.01	60.0	0.12	0.14	55.6	0.00	0.10
50.0	0.00	0.00	50.0	0.07	0.05	44.4	0.03	0.04
40.0	0.00	0.00	40.0	0.03	0.02	33.3	0.04	0.02
30.0	0.00	0.00	30.0	0.00	0.00	22.2	0.03	0.01
20.0	0.00	0.00	20.0	0.00	0.00	11.1	0.03	0.00
10.0	0.00	0.00	10.0	0.00	0.00	0.0	0.00	0.00
0.0	0.00	0.00	0.0	0.00	0.00			

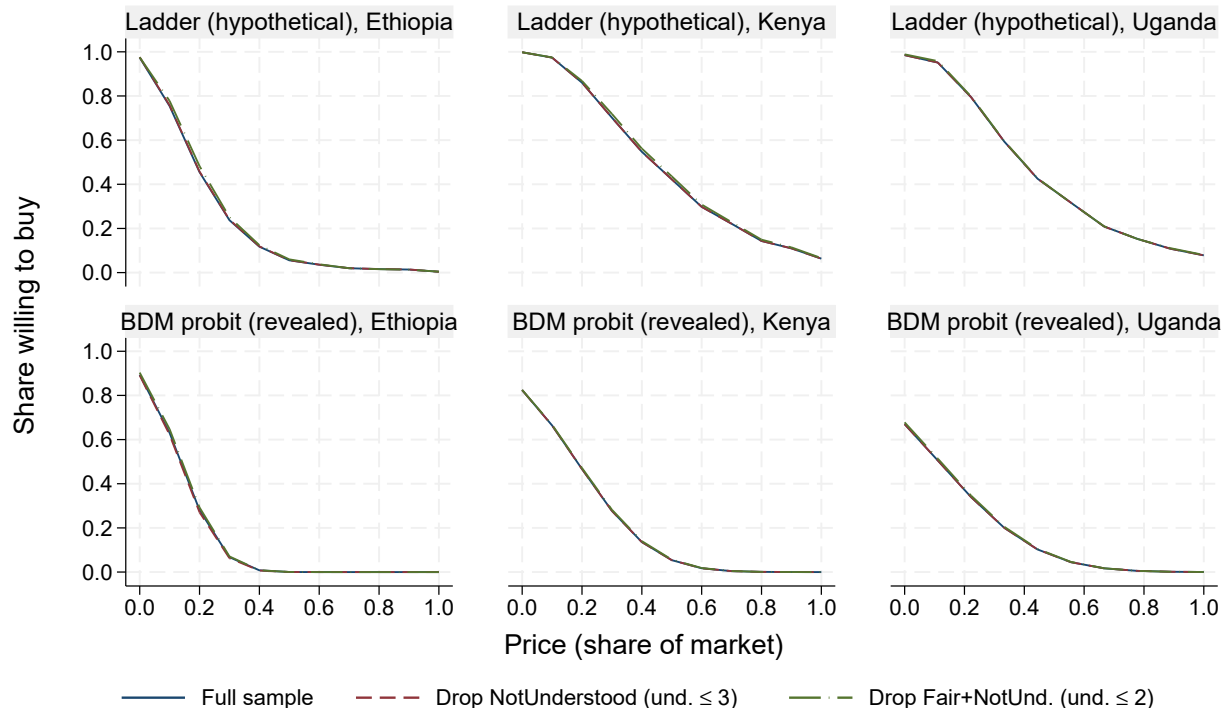
Notes: Each row reports the share of households willing to buy an EPC at a given subsidy level, estimated from the BDM binding offer. The implied EPC price is $(1 - \text{subsidy})$ as a share of market price. “Empirical” is the within-tier mean of $\mathbf{1}\{\text{buy}_i = 1\}$ among participants, scaled by the BDM participation share π_c (intention-to-treat; non-participants are non-buyers at every positive price). “Probit” is π_c times the predicted share from a per-country probit regression of $\mathbf{1}\{\text{buy}_i = 1\}$ on the drawn price p_i . The probit-implied price at 50% adoption (as a share of market price) is 0.095 in Ethiopia, 0.183 in Kenya, and 0.110 in Uganda, with delta-method 95% confidence intervals of $[0.071, 0.120]$, $[0.121, 0.245]$, and $[0.035, 0.184]$, respectively (standard errors clustered at the neighborhood level). Sample sizes match Figure 3. The ladder-only demand schedule (full sample, hypothetical) is reported in Table D4.

Table D6: BDM understanding by treatment arm and country

Understanding	All		Ethiopia		Kenya		Uganda	
	Control	Treat	Control	Treat	Control	Treat	Control	Treat
Very good	57.4	55.3	42.7	43.5	61.8	58.7	72.3	77.4
Good	37.3	38.0	47.3	46.0	34.6	37.1	26.6	17.9
Fair	5.0	6.3	9.2	9.2	3.5	4.2	1.1	4.8
Not understood	0.3	0.5	0.8	1.3	0.0	0.0	0.0	0.0
N	700	606	262	239	254	283	184	84

Notes: Column percentages of self-reported BDM-understanding category, by treatment arm and country. Categories: “Very good” (1), “Good” (2), “Fair” (3), “Not understood” (4).

Figure D2: Demand curves are robust to dropping low-understanding respondents



Notes: Demand schedules by country (columns) and elicitation method (rows). Top row: hypothetical ladder, within-tier empirical share of households stating they would purchase. Bottom row: BDM-revealed probit smoother from per-country probit regressions of $\mathbf{1}\{\text{buy}_i = 1\}$ on the drawn price p_i . Within each panel, the three curves restrict the sample to households reporting BDM understanding in (i) the full distribution, (ii) the top three categories (“Very good,” “Good,” or “Fair”; und. ≤ 3 , drops “Not understood”), and (iii) the top two categories (“Very good” or “Good”; und. ≤ 2 , drops “Fair” and “Not understood”). Implied prices at 50% adoption (P^{50} , share of market price) by country and restriction (full / und. ≤ 3 / und. ≤ 2): ladder Ethiopia 0.185 / 0.186 / 0.194; ladder Kenya 0.438 / 0.438 / 0.449; ladder Uganda 0.395 / 0.395 / 0.396; BDM probit Ethiopia 0.076 / 0.072 / 0.077; BDM probit Kenya 0.183 / 0.183 / 0.185; BDM probit Uganda never crosses 50% under any restriction. Sample sizes for the BDM probit match Figure 3 (per-protocol BDM sample); ladder sample is the full per-protocol set used in Table D4.

Table D7: The refusal gradient survives conditioning on the stated maximum

	Refused binding offer (=1)			
	(1)	(2)	(3)	(4)
Drawn price (% of market)	0.0105*** (0.0009)	0.0125*** (0.0011)	0.0088*** (0.0011)	0.0126*** (0.0012)
Stated maximum (% of market)		-0.0037*** (0.0010)		
Slack (maximum minus drawn price)			-0.0037*** (0.0010)	
Observations	340	340	340	340
R^2	0.326	0.354	0.354	0.429

Notes: Linear probability models of equation (4) on the at-risk sample (drawn price at or below stated ladder WTP). Column 1 is the unconditional gradient; column 2 controls for the stated ladder maximum M_i ; column 3 reparameterizes column 2 with the slack $M_i - P_i$; column 4 replaces the linear control with country-by- M -tier fixed effects. Columns 1–3 include country fixed effects (not shown). Standard errors, clustered at the neighborhood level (55 of the 63 neighborhoods appear in the at-risk sample), in parentheses; the text reports wild-cluster bootstrap p -values (Webb weights, 9,999 replications). The permutation test described in §4.2 (2,000 within-country reshuffles of M_i) provides the selection-only benchmark for column 1. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D8: Overstatement is weakly predictable, and its predictors indicate a generic response style

	Overstated (=1)
Respondent education (level)	-0.031** (0.015)
Respondent female	0.017 (0.065)
Female household head	-0.087 (0.059)
Asset index (PCA)	-0.017 (0.016)
Monthly expenditure (asinh USD)	0.020 (0.034)
Main fuel: charcoal	0.109 (0.087)
Main fuel: LPG	0.056 (0.087)
Household size	-0.016 (0.012)
Familiar with EPCs	-0.041 (0.056)
Jelly-ladder WTP (% of jelly price)	0.003*** (0.001)
BDM comprehension score	0.067 (0.044)
Perceives electricity most expensive	0.017 (0.053)
Observations	340
Pseudo R^2	0.081
Correctly classified (% , $\hat{p} \geq 0.5$)	66.5
Base rate (% overstaters)	62.1

Notes: Average marginal effects from the probit in equation (5), estimated on the at-risk sample (drawn price at or below stated ladder WTP). The outcome equals 1 if the household refused the binding offer. Country fixed effects included (not shown). Standard errors, clustered at the neighborhood level, in parentheses. The asset index is the first principal component of the household asset battery. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D9: Sensitivity of WTP correlates to BDM consistency

	(1) Full	(2) Drop overest.	(3) Drop both	(4) Cap WTP	(5) Drop+High und.
Respondent age	-0.075 (0.066)	-0.138** (0.055)	-0.137** (0.055)	-0.092* (0.055)	-0.152** (0.061)
HH head female	-1.021 (1.365)	0.658 (1.295)	0.749 (1.292)	-0.423 (1.208)	0.029 (1.354)
Household size	0.210 (0.308)	-0.029 (0.282)	-0.024 (0.282)	0.002 (0.256)	-0.074 (0.290)
Primary cook age	-0.024 (0.073)	0.110* (0.062)	0.109* (0.062)	0.063 (0.060)	0.121* (0.068)
Primary cook is hired help	0.351 (2.913)	3.763 (2.444)	3.763 (2.444)	2.803 (2.446)	4.314* (2.589)
Monthly expenditure (asinh, USD)	4.026*** (0.895)	3.880*** (0.796)	3.870*** (0.796)	3.965*** (0.745)	3.867*** (0.821)
Familiar with EPC	2.330 (1.549)	1.508 (1.428)	1.569 (1.427)	1.082 (1.328)	1.020 (1.472)
Main cooking fuel: charcoal	-4.399*** (1.684)	-5.427*** (1.557)	-5.518*** (1.555)	-4.443*** (1.517)	-5.147*** (1.645)
Main cooking fuel: LPG	4.897** (2.187)	3.739* (1.996)	3.744* (1.995)	5.435*** (1.815)	3.785* (2.059)
Grid is main elec. source	4.640* (2.759)	2.439 (2.542)	2.367 (2.542)	3.355 (2.381)	2.501 (2.569)
Typical outage hrs/week	-0.001 (0.045)	0.013 (0.043)	0.011 (0.043)	0.004 (0.041)	0.001 (0.045)
Constant	-6.037 (6.240)	-6.537 (5.748)	-6.447 (5.747)	-7.694 (5.269)	-5.017 (6.016)
Observations	1304	1093	1091	1304	1020
R^2	0.248	0.222	0.223	0.222	0.210

Notes: OLS estimates of the pooled correlates regression (equation 7; pooled with country fixed effects, suppressed). Outcome is WTP as share of market price (0–100). Heteroskedasticity-robust standard errors in parentheses, matching the parent specification in Table C1 column 5. “Drop overest.” drops respondents who could buy at the drawn price per their BDM but elected not to (`can_buy=1`, `buy_EPC=0`). “Drop both” additionally drops respondents who bought even though their stated WTP was below the drawn price. “Cap WTP” replaces stated WTP with $\max(0, \text{drawn price} - 10)$ for overestimating respondents instead of dropping them. “Drop+High und.” drops overestimators and restricts to respondents reporting “Very good” or “Good” BDM understanding. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D10: Heterogeneous information-treatment effects on WTP

	(1) High exp.	(2) Charcoal	(3) LPG	(4) Elec. exp.	(5) Know EPC
Info treatment=1	-0.707 (1.799)	-1.387 (1.484)	-0.460 (1.553)	-0.398 (1.816)	-0.082 (1.536)
High exp. (within country)=1	6.567*** (1.842)				
Info treatment=1 × High exp. (within country)=1	-0.529 (2.632)				
Respondent female	0.107 (1.891)	-0.873 (1.920)	-0.822 (1.921)	-0.846 (1.924)	-0.599 (1.926)
Household size	0.283 (0.308)	0.634** (0.305)	0.643** (0.305)	0.642** (0.306)	0.639** (0.306)
Main cooking fuel: charcoal	-4.678*** (1.716)		-5.096*** (1.728)	-5.131*** (1.732)	-5.003*** (1.724)
Main cooking fuel: LPG	5.256** (2.167)	6.133*** (2.194)		6.100*** (2.192)	5.991*** (2.187)
Typical outage hrs/week	-0.011 (0.045)	-0.012 (0.045)	-0.013 (0.045)	-0.013 (0.045)	-0.011 (0.045)
Main cooking fuel: charcoal=1		-6.353*** (2.297)			
Info treatment=1 × Main cooking fuel: charcoal=1		2.850 (3.271)			
Main cooking fuel: LPG=1			6.481** (2.640)		
Info treatment=1 × Main cooking fuel: LPG=1			-0.746 (2.864)		
Elec. most expensive=1				-0.131 (1.816)	
Info treatment=1 × Elec. most expensive=1				-0.760 (2.643)	
Familiar with EPC=1					4.041* (2.082)
Info treatment=1 × Familiar with EPC=1					-1.749 (2.930)
Constant	13.468*** (2.755)	15.996*** (2.650)	15.502*** (2.672)	15.759*** (2.814)	14.487*** (2.651)
Observations	1304	1304	1304	1304	1304
R ²	0.239	0.227	0.227	0.227	0.230

Notes: OLS estimates of the information treatment on WTP (share of market price, 0–100), interacted with the dimension named in the column header. Country fixed effects in all columns. Heteroskedasticity-robust standard errors. “High exp.” is an indicator for above-median total monthly expenditure within the household’s country. “Elec. exp.” is an indicator for the respondent perceiving electricity as the most expensive cooking fuel. “Know EPC” is an indicator for baseline familiarity with electric pressure cookers. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D11: Gender heterogeneity of the information-treatment effect on WTP

	Respondent female				HH head female			
	(1) Pooled	(2) ET	(3) KE	(4) UG	(5) Pooled	(6) ET	(7) KE	(8) UG
Info treatment=1	4.431 (3.387)	3.119 (3.838)	-2.448 (7.807)	14.966 (11.109)	-0.867 (1.652)	-2.625 (1.531)	-2.564 (2.914)	9.091 (5.544)
Respondent female=1	2.276 (2.141)	2.023 (2.156)	-2.509 (7.678)	4.538 (4.018)				
Info treatment=1 × Respondent female=1	-5.915* (3.445)	-5.121 (4.113)	0.281 (6.988)	-12.720 (12.880)				
Respondent female					0.256 (2.205)	-0.524 (2.307)	-0.898 (5.242)	2.402 (5.333)
Household size	0.675** (0.324)	0.572 (0.399)	0.832 (0.897)	0.678 (0.510)	0.550 (0.336)	0.515 (0.395)	0.633 (0.965)	0.507 (0.494)
Main cooking fuel: charcoal	-5.019*** (1.887)	-7.414*** (1.320)	-2.277 (7.178)	-3.738 (5.080)	-5.093*** (1.812)	-7.332*** (1.209)	-2.426 (7.080)	-4.864 (5.095)
Main cooking fuel: LPG	6.272** (2.460)	1.472 (3.515)	9.514*** (2.948)	1.942 (7.606)	5.866** (2.488)	1.395 (3.604)	9.084*** (2.949)	0.198 (7.455)
Typical outage hrs/week	-0.013 (0.048)	-0.007 (0.043)	-0.048 (0.125)	0.131 (0.265)	-0.009 (0.048)	-0.009 (0.043)	-0.031 (0.125)	0.126 (0.262)
HH head female=1					-3.084 (2.334)	-2.401 (2.325)	-4.715 (5.542)	-1.243 (4.384)
Info treatment=1 × HH head female=1					0.557 (2.585)	3.766 (3.520)	1.674 (5.247)	-10.135 (5.972)
Constant	12.678*** (3.389)	14.273*** (3.562)	36.037*** (8.100)	33.555*** (6.880)	16.312*** (3.244)	17.748*** (3.252)	37.068*** (6.632)	37.857*** (6.417)
Observations	1304	499	537	268	1304	499	537	268
R ²	0.228	0.037	0.028	0.020	0.229	0.037	0.032	0.027

Notes: OLS estimates of the information treatment on WTP (share of market price, 0–100), interacted with the respondent's gender (columns 1–4) and the household head's gender (columns 5–8). Columns 1 and 5 are pooled with country fixed effects; columns 2–4 and 6–8 are estimated within country. Controls in all columns: household size, main cooking fuel charcoal, main cooking fuel LPG, typical weekly outage hours (winsorized at country-specific p95). Columns 1–4 omit respondent-female from the control set (interacted variable); columns 5–8 retain respondent-female as a control. Cluster-robust standard errors at the neighborhood level. Cluster counts: pooled 63, Ethiopia 13, Kenya 26, Uganda 24; small-cluster asymptotics caution applies for the by-country columns. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D12: Sensitivity of pooled information-treatment effect to BDM consistency

	(1) Full	(2) Drop overest.	(3) Drop both	(4) Cap WTP	(5) Drop+High und.
Info treatment	-0.744 (1.300)	-1.047 (1.256)	-1.214 (1.248)	-0.320 (1.116)	-0.771 (1.377)
Respondent female	-0.838 (2.009)	-1.284 (1.637)	-1.352 (1.663)	-1.330 (1.696)	-0.806 (1.624)
Household size	0.648** (0.321)	0.330 (0.305)	0.331 (0.305)	0.414 (0.262)	0.289 (0.311)
Main cooking fuel: charcoal	-5.099*** (1.862)	-5.898*** (1.735)	-5.978*** (1.721)	-5.123*** (1.718)	-5.287*** (1.948)
Main cooking fuel: LPG	6.107** (2.475)	4.466* (2.263)	4.473* (2.257)	6.445*** (2.258)	4.511* (2.429)
Typical outage hrs/week	-0.013 (0.048)	0.009 (0.044)	0.007 (0.044)	0.000 (0.045)	-0.007 (0.047)
Constant	15.634*** (3.037)	16.326*** (2.757)	16.521*** (2.768)	15.571*** (2.535)	17.081*** (2.929)
Observations	1304	1093	1091	1304	1020
R^2	0.227	0.200	0.201	0.200	0.187

Notes: OLS estimates of the information treatment on WTP (share of market price, 0–100). All columns cluster at the neighborhood level. “Drop overest.” drops respondents who could buy at the drawn price per their BDM but elected not to (`can_buy=1`, `buy_EPC=0`; 318 households). “Drop both” additionally drops respondents who bought even though their stated WTP was below the drawn price (4 households). “Cap WTP” replaces stated WTP with $\max(0, \text{drawn price} - 10)$ for overestimating respondents instead of dropping them. “Drop+High und.” drops overestimators and restricts to respondents reporting “Very good” or “Good” BDM understanding. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D13: Per-protocol gating matters only in Ethiopia: ITT and per-protocol subsidy targets

	BDM	Per-protocol		ITT		
	particip. share	P50 price (% mkt)	Subsidy (%)	P50 price (% mkt)	Subsidy (%)	95% CI (P50)
Ethiopia	0.772	13.6	86.4	9.5	90.5	[7.0, 12.1]
Kenya	0.998	18.4	81.6	18.3	81.7	[12.0, 24.6]
Uganda	0.985	11.4	88.6	11.0	89.0	[3.7, 18.2]

Notes: Per-protocol columns fit the binding-choice probit on BDM participants only. ITT columns code non-participants as never-buyers at every positive price: country-wide demand is $D_c^{\text{ITT}}(p) = \pi_c \Phi(\hat{a}_c + \hat{b}_c p)$, where π_c is the BDM participation share (387/501 in Ethiopia, 536/537 in Kenya, 264/268 in Uganda). All 114 Ethiopian non-participants stated a ladder maximum of zero, so the never-buyer coding follows from their stated choice; the 1 Kenyan and 4 Ugandan non-participants are a worst-case bound and are numerically negligible. P^{50} is the price (percent of market) at which country-wide predicted adoption reaches 50%; the subsidy is 100 minus P^{50} . Confidence intervals use the delta method with $t(G - 1)$ critical values, where G is the country cluster count. The ITT estimates are the paper's headline subsidy figures (§5.1); the per-protocol estimates are robustness.

Table D14: The EPC repays its price within months, but the payback period does not predict refusal

Panel A: Payback economics by country			
	Ethiopia	Kenya	Uganda
Payback at drawn price, p25 (months)	2.1	2.3	0.8
Payback at drawn price, median (months)	3.9	4.8	2.1
Payback at drawn price, p75 (months)	7.5	9.4	4.4
Payback at market price, median (months)	7.8	7.1	4.7
Break-even meals at market price, median	467	495	253
Households with non-positive savings (%)	0.0	8.4	15.1
Market price / monthly expenditure (%)	66.9	50.4	47.0
Households with a draw and positive savings	387	491	223
Panel B: Refusal of the binding offer (probit AMEs)			
Drawn price (% of market price)	0.010*** (0.001)		
ln(payback months at drawn price)	0.003 (0.021)		
Observations	272		
Pseudo R^2	0.204		

Notes: Panel A: payback periods divide the EPC price (drawn or market) by predicted monthly fuel-cost savings, computed from country tariffs prevailing at enumeration, 0.5 kWh per meal, and each household's reported fuel expenditure per meal (winsorized at country-specific 5th/95th percentiles). Payback statistics condition on households with a drawn price and positive predicted savings; break-even meals, non-positive-savings shares, and expenditure shares use all households with the required data. Panel B: average marginal effects from a probit of refusal on the drawn price and the log household-specific payback period, estimated on the at-risk sample with country fixed effects (not shown); standard errors, clustered at the neighborhood level, in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D15: Ladder demand schedule under consistency adjustments: Ethiopia

Subsidy (%)	Price (share)	Full sample	Drop overest.	Drop both	Cap (-10pp)	Cap (exact)	Drop+ HighUnd
100.0	0.000	0.97	0.97	0.97	1.00	1.00	0.97
90.0	0.100	0.75	0.74	0.74	0.73	0.75	0.76
80.0	0.200	0.46	0.43	0.43	0.43	0.44	0.46
70.0	0.300	0.24	0.22	0.22	0.10	0.10	0.23
60.0	0.400	0.12	0.10	0.10	0.10	0.10	0.10
50.0	0.500	0.06	0.04	0.04	0.04	0.05	0.05
40.0	0.600	0.04	0.03	0.03	0.02	0.02	0.03
30.0	0.700	0.02	0.01	0.01	0.02	0.02	0.02
20.0	0.800	0.02	0.01	0.01	0.01	0.01	0.01
10.0	0.900	0.01	0.01	0.01	0.01	0.01	0.01
0.0	1.000	0.00	0.00	0.00	0.00	0.00	0.00

Notes: Share of households willing to buy an EPC at each ladder price under alternative consistency adjustments. “Full sample” uses the unadjusted stated ladder. “Drop overest.” drops overestimators (households that stated they could buy at the drawn price but refused the binding BDM offer); “Drop both” additionally drops the reverse inconsistency (households that bought even though their stated maximum was below the drawn price). “Cap (−10pp)” and “Cap (exact)” retain overestimators but cap their stated WTP at the drawn price minus 10 percentage points and at exactly the drawn price, respectively. “Drop+HighUnd” drops overestimators and restricts to households with high BDM comprehension scores. See Appendix B for construction details.

Table D16: Ladder demand schedule under consistency adjustments: Kenya

Subsidy (%)	Price (share)	Full sample	Drop overest.	Drop both	Cap (-10pp)	Cap (exact)	Drop+ HighUnd
100.0	0.000	1.00	1.00	1.00	1.00	1.00	1.00
90.0	0.100	0.97	0.97	0.97	0.96	0.97	0.97
80.0	0.200	0.86	0.83	0.83	0.83	0.85	0.84
70.0	0.300	0.70	0.64	0.64	0.49	0.51	0.65
60.0	0.400	0.55	0.44	0.44	0.49	0.51	0.46
50.0	0.500	0.42	0.30	0.30	0.33	0.38	0.31
40.0	0.600	0.30	0.18	0.18	0.13	0.14	0.19
30.0	0.700	0.22	0.11	0.11	0.13	0.14	0.12
20.0	0.800	0.14	0.05	0.05	0.07	0.08	0.05
10.0	0.900	0.11	0.03	0.03	0.03	0.05	0.03
0.0	1.000	0.06	0.01	0.01	0.01	0.01	0.01

Notes: See notes to Table D15. Columns apply the same consistency adjustments to the Kenya ladder sample.

Table D17: Ladder demand schedule under consistency adjustments: Uganda

Subsidy (%)	Price (share)	Full sample	Drop overest.	Drop both	Cap (-10pp)	Cap (exact)	Drop+ HighUnd
100.0	0.000	0.99	0.98	0.98	1.00	1.00	0.99
88.9	0.111	0.95	0.94	0.94	0.74	0.74	0.94
77.8	0.222	0.79	0.74	0.74	0.53	0.53	0.75
66.7	0.333	0.59	0.50	0.50	0.34	0.34	0.50
55.6	0.444	0.43	0.33	0.33	0.22	0.22	0.33
44.4	0.556	0.32	0.23	0.23	0.21	0.22	0.23
33.3	0.667	0.21	0.15	0.14	0.09	0.10	0.14
22.2	0.778	0.15	0.09	0.08	0.09	0.09	0.09
11.1	0.889	0.11	0.05	0.04	0.03	0.05	0.05
0.0	1.000	0.08	0.03	0.03	0.03	0.03	0.04

Notes: See notes to Table D15. Columns apply the same consistency adjustments to the Uganda ladder sample.

Table D18: Sensitivity of pooled information-treatment effect to BDM understanding

	(1) Full	(2) Drop NU	(3) Drop NU+Fair	(4) Full (cl)	(5) Drop NU (cl)	(6) Drop NU+Fair (cl)
Info treatment	-0.744 (1.339)	-0.688 (1.344)	-0.345 (1.396)	-0.744 (1.300)	-0.688 (1.298)	-0.345 (1.401)
Respondent female	-0.838 (1.922)	-0.768 (1.923)	-0.497 (1.985)	-0.838 (2.009)	-0.768 (2.004)	-0.497 (1.988)
Household size	0.648** (0.305)	0.643** (0.307)	0.636** (0.317)	0.648** (0.321)	0.643** (0.320)	0.636* (0.328)
Main cooking fuel: charcoal	-5.099*** (1.728)	-5.084*** (1.737)	-4.304** (1.823)	-5.099*** (1.862)	-5.084*** (1.887)	-4.304** (2.036)
Main cooking fuel: LPG	6.107*** (2.190)	6.087*** (2.192)	6.280*** (2.242)	6.107** (2.475)	6.087** (2.479)	6.280** (2.566)
Typical outage hrs/week	-0.013 (0.045)	-0.015 (0.046)	-0.028 (0.047)	-0.013 (0.048)	-0.015 (0.048)	-0.028 (0.050)
Constant	15.634*** (2.641)	15.717*** (2.652)	16.112*** (2.717)	15.634*** (3.037)	15.717*** (3.056)	16.112*** (3.151)
Observations	1304	1299	1226	1304	1299	1226
R ²	0.227	0.224	0.219	0.227	0.224	0.219

Notes: OLS estimates of the information treatment on WTP (share of market price, 0–100). Columns 1–3 use heteroskedasticity-robust standard errors; columns 4–6 cluster at the neighborhood level (63 clusters). “Drop NU” restricts to respondents with self-reported BDM understanding above “Not understood.” “Drop NU+Fair” restricts further to respondents reporting “Very good” or “Good” understanding. Controls: respondent female, household size, charcoal use, LPG use, country FE. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D19: Practice-BDM (jelly) demand schedule

Subsidy (%)	Price (share of full)	Share willing to buy			
		All	Ethiopia	Kenya	Uganda
100	0.00	0.99	0.98	1.00	0.99
90	0.10	0.96	0.91	1.00	0.98
80	0.20	0.91	0.80	1.00	0.93
70	0.30	0.84	0.66	1.00	0.84
60	0.40	0.73	0.47	0.99	0.71
50	0.50	0.67	0.39	0.96	0.62
40	0.60	0.58	0.35	0.94	0.28
30	0.70	0.50	0.26	0.85	0.24
20	0.80	0.45	0.21	0.80	0.20
10	0.90	0.38	0.16	0.70	0.15

Notes: Share of households indicating they would buy a jar of jelly at each price point in the practice BDM ladder. Practice ladder spans 0%–90% of full price in 10pp steps.

Table D20: Practice-BDM WTP predicts EPC WTP

	(1) Bivariate	(2) + Country FE	(3) + Controls + Treat	(4) + Interaction
Jelly BDM WTP	0.352*** (0.021)	0.229*** (0.025)	0.220*** (0.026)	0.237*** (0.033)
Info treatment			-1.329 (1.317)	
Respondent female			-1.187 (1.871)	-1.192 (1.868)
Household size			0.428 (0.302)	0.419 (0.302)
Main cooking fuel: charcoal			-4.574*** (1.684)	-4.553*** (1.681)
Main cooking fuel: LPG			5.378** (2.114)	5.407** (2.111)
Typical outage hrs/week			-0.019 (0.045)	-0.017 (0.045)
Info treatment=1				0.815 (2.345)
Info treatment=1 $\times$ Jelly BDM WTP				-0.035 (0.040)
Constant	11.506*** (1.220)	7.474*** (1.167)	8.154*** (2.604)	7.226*** (2.717)
Observations	1306	1306	1304	1304
R^2	0.154	0.252	0.262	0.262

Notes: OLS estimates of EPC WTP on practice-BDM (jelly) WTP. Both WTP measures are on the 0–100 share-of-full-price scale (jelly ladder is 0–90 in 10pp steps; EPC ladder is 0–100 in roughly 10pp steps). Column 2 adds country fixed effects; column 3 adds the information-treatment indicator and the pooled control set; column 4 interacts jelly WTP with the information treatment. Heteroskedasticity-robust standard errors. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.